MSE 281: Modelling of Mechatronic Systems

Lab 4, 5 and Project: Forced Response Analysis of a Mass-Spring-Damper

Group 9:

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Executive Summary

In this lab we will extend our study of Mass-Spring-Damping (MSD) systems to forced vibration and frequency response. Using OSCAR with different springs, step, and sinusoidal inputs we determine actuator gain, spring stiffness, damping, friction, and system mass from both time domain and steady state responses. We then use the vertical beam on OSCAR to model a flexible structure under base excitation, estimating its effective mass, stiffness, damping ratio, and natural frequency from its step response. With these parameters we build an equivalent RLC circuit model that matches the second order differential equation of the mechanical MSD system and compare the simulated electrical response to the measured motion. From our results we see how heavy damping and friction prevent a clear resonance peak and cause differences between step response and frequency response estimates, while still showing the same second order vibration behavior. Overall this lab has strengthened the connection between differential equations, real vibration behaviour, and their electrical analogs, while highlighting imperfections in real systems.

Introduction

In Lab 4 we have a focus on determining properties of a mass-spring-damping system (MSD). Through the experiment we characterize mass (m), stiffness (k), and damping (c) of an MSD system named "OSCAR". We use two different inputs to find these parameters: step input and sinusoidal input. In addition to discovering those parameters we calculate static friction and the second order transfer function of a vertical beam attached to OSCAR. Moving into our project our goal is to model the MSD system in lab 4 as an electrical circuit. This works as the second order differential equation to model an MSD system ($m\ddot{x} + c\dot{x} + kx = F$) is equal to the second order differential equation of an electric circuit (RLC: $L\ddot{V}_c + R\dot{V}_c + \frac{1}{C}V_c = \frac{V_{in}}{C}$). We will simulate the mechanical system using electrical components in an RLC circuit. We take the parameters (m, k, c) from the MSD system of lab 4 and change them to inductance (L), resistance (R) and capacitance (C) to model an equivalent circuit.

Description of Experiments

Apparatus and Materials:

Lab 4

OSCAR

- Weights
- Springs
- Computer Software

Project

- Resistor
- Inductor
- Capacitor
- Oscilloscope
- Breadboard
- Wires

Procedure:

Lab 4

1. Setup OSCAR and software
2. Start Part 1a
3. Start with no springs an 0kg
4. Change α until the system barely moves
5. Add 1kg repeat step 4
6. Add 2kg repeat step 4
7. Keep 2kg
8. Add red spring then repeat step 4
9. Remove red spring add green spring then repeat step 4
10. Remove green spring add red spring then repeat step 4
11. Compute calculations
12. Start Part 1b
13. Use α from part a
14. Keep 2kg add red spring
15. Run sinusoid input at a frequency of 5rad/s, 10 rad/s, and 25rad/s
16. Measure steady-state displacement amplitude and phase shift
17. Repeat steps 14-16 for the green and white spring
18. Compute calculations for Part 2

Project

1. Gather components needed for the circuits
2. Part 1
3. Build OSCAR circuit
4. Test circuit with DC input with f = 100Hz, Amp. = 1 – 2 Vpp, offset = 0 V
5. Record data for phase and amplitude
6. Repeat steps 3 – 5 for beam circuit, and OSCAR and beam circuit together.
7. Part 2
8. Build OSCAR circuit again
9. You are measuring phase and amplitude again
10. Test with sine input and f = 200, 500, 1k, 1.5k, 2k, 3k, and 5k Hz
11. Build Beam circuit
12. You are measuring phase and amplitude again
13. Test with sine input and f = 1k, 3k, 5k, 7k, 8k, 10k, 15k Hz
14. Build OSCAR and beam circuit together
15. You are measuring phase and amplitude again
16. Test with sine input and f = 500, 1k, 3k, 7k, 8k, 10k Hz
17. Clean up material

Calculation Method

Steady state (x_{ss})	The displacement when oscillations have stopped
Precent Overshoot (PO)	$PO = \frac{\lvert x_{peak}-x_{ss}\rvert}{\lvert x_{start}-x_{ss}\rvert} * 100\%$(There is no precent overshoot as the system is overdamped)
Rise time	Time between 10% and 90% of response is reached
Settle time	Time after release until the position is permanently within ±2% of steady state
Damping Frequency (ω_d)	$\omega_d = \frac{2\pi}{T_d}$
Logarithmic Decrement (δ)	$\delta = \frac{1}{n}\ln\left(\frac{x(t)}{x(t+nT_d)}\right)$
Damping Ratio (ζ)	$\zeta = \frac{\delta}{\sqrt{\pi^2+\delta^2}}$

Natural Frequency (ω_n) (Experimental) $\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$

Natural Frequency (ω_n) (Theoretical) $\omega_n = \sqrt{\frac{k}{m}}$

Error in ω_n $= \frac{|\omega_{n\,exp} - \omega_{ntheo}|}{\omega_{nthe}} * 100\%$

Lab 4

Part 1 A

Assuming the stepper motor voltage is linearly proportional to force apply to cart. We define the gain as y such that $\alpha y = F$.

There for we can solve for y using:

$$\frac{\alpha_f y + \alpha_r y}{2} = k\Delta$$

$$y = \frac{2k\Delta}{\alpha_f + \alpha_r}$$

Then use y to solve for force of friction.

$$F_{fr} = \frac{y(\alpha_f - \alpha_r)}{2}$$

Using this we get values for each of the springs tested using this matlab code.

```
a=[0.65 0.45;0.5 0.42; 0.45 0.4]
k=[1226 700 263]'
y=2*k*0.001./(a(:,1)+a(:,2))
Ffr=y.*(a(:,1)+a(:,2))/2
g = 9.81;
```

Spring	Red	Green	White	Avg
y = (N/V)	2.2291	1.5217	0.6188	1.46
F_{fr} = (N)	1.2260	0.7000	0.2630	0.7297

Using the mass from lab 1 (2.15kg) to solve for coefficient of friction (μ):

$$F_{fr} = \mu F_N = \mu mg$$

$$\mu = \frac{F_{fr}}{mg} = \frac{0.7297}{2.15 \cdot 9.81} = 0.0346$$

Using the amplitude and frequency from part B the RMS velocity is 0.005794m/s

$$v_{RMS} = \frac{(X \cdot \omega)}{\sqrt{2}}$$

$$F_{fr} = c\dot{x}$$

Using this we can get a rough approximation for c = 126 Ns/m.

Part 1B

Using frequence response we have the equations:

$$|H(j\omega)| = \frac{X}{F_0} = \frac{1}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

$$\varphi = \tan^{-1}\left(\frac{c\omega}{k - m\omega^2}\right)$$

Solving for c and m we get the equations:

$$c = \frac{\tan(\varphi)\,(k - m\omega^2)}{\omega}$$

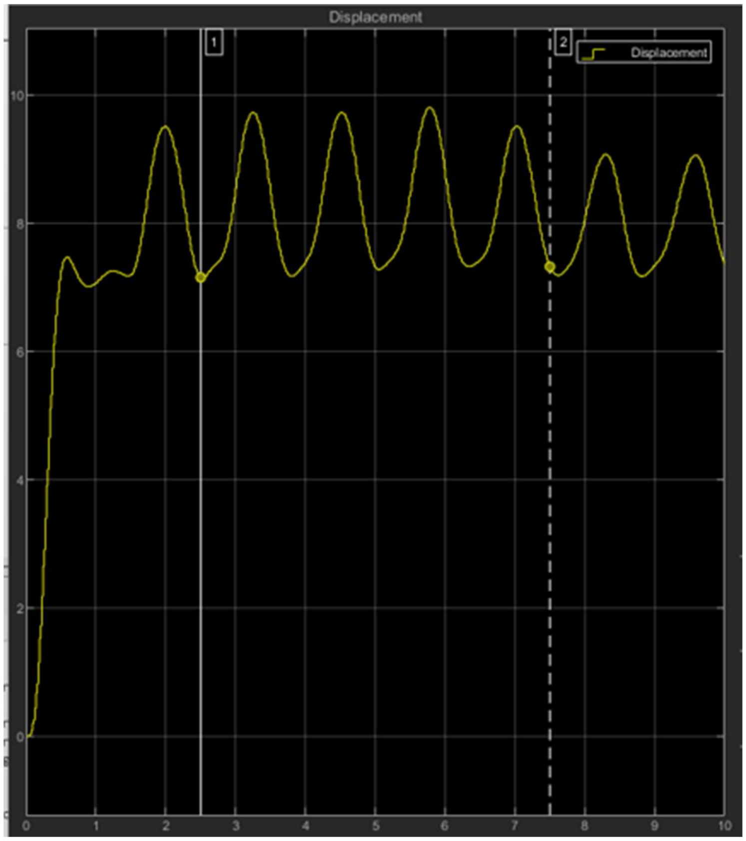
$$m = \frac{\left(k - \cos\varphi\,\frac{F_0}{X}\right)}{\omega^2}$$

$$c = \frac{\sqrt{\frac{F_0}{X} - (k - m\omega^2)}}{\omega}$$

Estimating for F_0 from the from the Input Force Graphs we should have gathered in Lab4:

$$A_{pk-pk} = 2F_0$$

Since F_0 (the driving force of the motor) should be consistent throughout the experiment regardless of frequency we can collect it from any of our sinusoidal input experiments.



From this graph we can read off that the $A_{pk\text{-}pk}$ is equal to approximately 2N. Giving us an F_0 = 1N

Using this matlab code we solve for *m* and *c*:

```
phi = [22.5 45 90; 60 60 150; 90 90 210];
A=[2 3 3; 1 1.5 2; 0.5 0.5 1.5]/2000; %X
w=[5 10 25];
F_0 = [1 1 1];
m = (k - cos(phi*pi/180).* F_0 ./ A) ./ (w.^2)
mavg = sum(m(:))/9;
c=sqrt(((F_0.^2./ A.^2)-(k-mavg.*w.^2).^2./w.^2))
cavg = sum(c(:))/9;
```

M (kg)	Red	Green	White
5 rad/s	12.0848	7.5460	1.9616
10 rad/s	-12.0000	0.3333	2.5056
25 rad/s	10.5200	2.6300	2.2683

m_{avg} = 3.0735 kg

c (Ns/m)	Red	Green	White
5 rad/s	973.3	660.3	666.1
10 rad/s	1996.1	1332.8	998.8
25 rad/s	3999.8	4000.0	1331.7

c_{avg} = 1773.21 Ns/m

We can solve for the nature frequency (ω_n) and damping ratio (ζ) using c, k, and m with the equations below.

$$\omega_n = \sqrt{\frac{k}{m}} \qquad \zeta = \frac{c}{2m\omega_n}$$

	m (kg)	k (N/m)	c (Ns/m)	ω_n (rad/s)	ζ
Part A	2.15	1226	126	23.9	1.23
Part B	3.0735	1226	1773.21	19.97	14.45

Part 2A

Calculating stainless steel beam properties

t =1mm, L = 300mm, W = 14mm

E = 210 GPa, ρ = 7850 kg/m^3

m = 7850 kg/m^3 * 0.001m * 0.300m * 0.014m = 0.03297kg

Since the mass is distributed and does not move in unison we use the effective mass in our calculations.

$$m_{eff} = m \cdot 0.236 = 7.78 \cdot 10^{-3} kg$$

The stiffness of the bar is calculated using the equation for a cantilever bean shown below.

$$k_s = \frac{3EI}{L^3}, \qquad I = \frac{wt^3}{12}$$

$$k_s = 27.2\ \frac{N}{m}$$

$$c = m_{eff} \cdot 2\zeta \sqrt{\frac{k}{m_{eff}}} = 4.60 \cdot 10^{-3}\ \frac{Ns}{m}$$

The equation for a mass, spring, dampener system and an LCR circuit are very similar. We can model a mass, spring, dampener system with an LCR circuit where *m = L, c = R,* and *k = 1/C.*

$$m\ddot{x} + c\dot{x} + kx = F, \qquad L\ddot{V_c} + R\dot{V_c} + \frac{1}{C}V_c = \frac{V_{in}}{C}$$

Equivalent circuit element values:

	Inductor (H)	Resistor (Ω)	Capacitor (F)
Beam	0.00778	0.0046	0.03676
Oscar	3.0735	1773.21	0.0692041522

Values Collected and Calculated

Mass	0 kg	1kg	2kg	2kg	2kg	2kg
Spring	None	None	None	Red	Green	White
$\alpha_{forward}$	0.60	0.62	0.62	0.65	0.42	0.45
$\alpha_{reverse}$	0.44	0.46	0.54	0.45	0.5	0.4

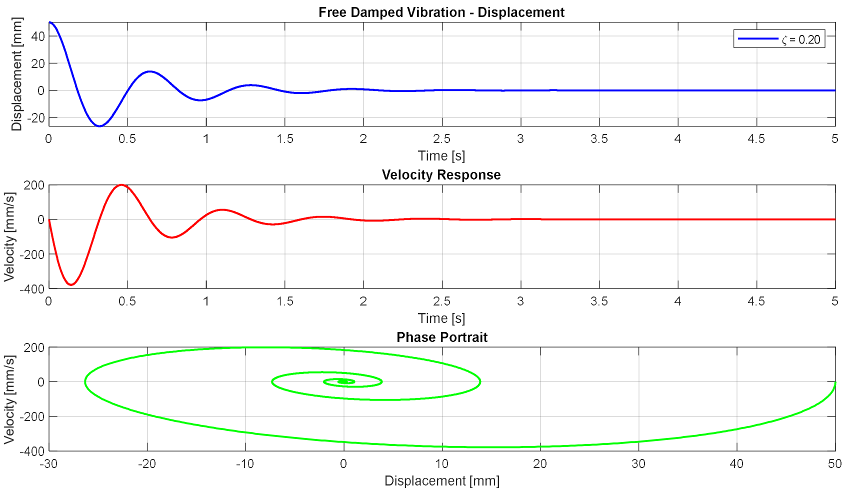
$X_{pk\text{-}pk}$ (mm)	Red	Green	White
5 rad/s	2	2	3
10 rad/s	1	1.5	2
25 rad/s	0.5	0.5	1.5

φ (degrees)	Red	Green	White
5 rad/s	22.5	45	90
10 rad/s	60	60	150
25 rad/s	90	90	210

Lab 5

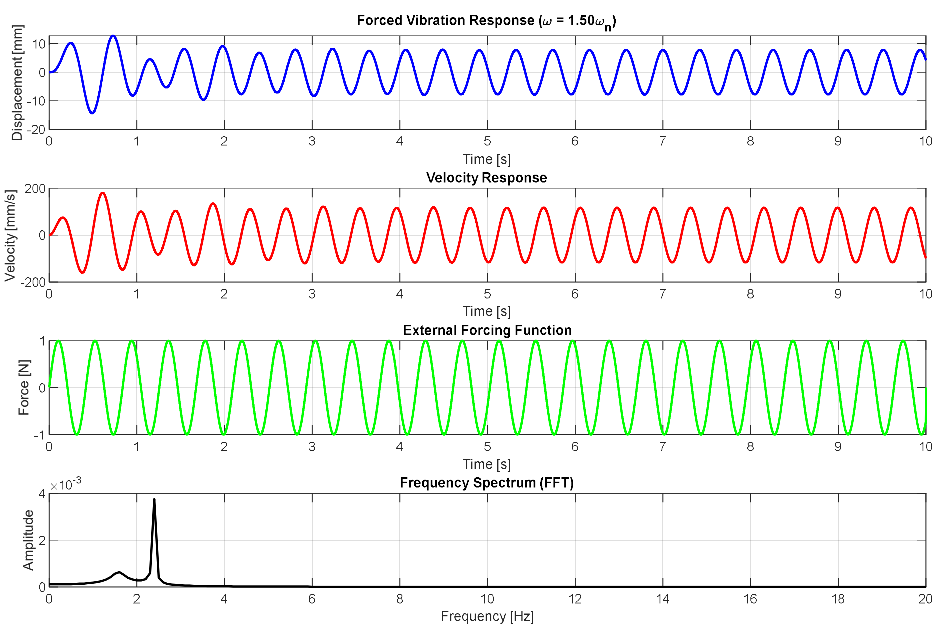
Part 1: Free Damped Vibration

Using the following system parameters; m = 1kg, k = 100N/m, ζ=0.2, and using a dt of 0.001s and end time of 5s. Setting initial conditions to x(1) = 0.05, v(1) =0, and x(2)=x(1)+v(1)*dt. We use central difference method to numerically simulate position, velocity, and acceleration.



Part 2: Forced Vibration (Harmonic Excitation)

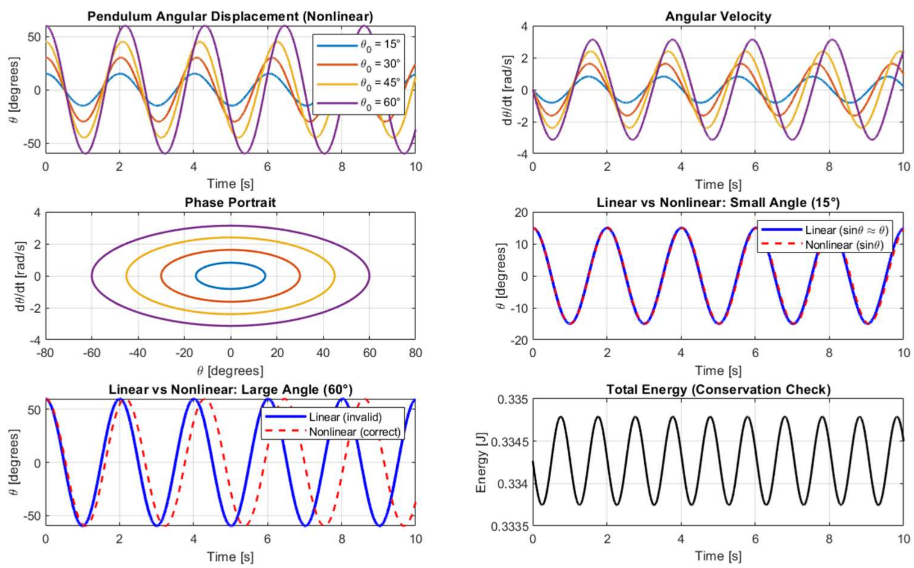
We use the same system parameter as before except using ζ=0. We apply a sinusoidal forces amplitude of 1N at a frequency of 1.5 times the natural frequency. Using the same numerical simulation method as before. Setting initial conditions to be 0 and an end time of 10s we see the following response.



For the bottom plot we used fast Fourier transform (FFT). The large peak is at 2.39Hz which is the excitation frequency which was expected steady state response. The second peak is at 1.59Hz, the natural frequency. This smaller spike is due to the transient response and will be less noticeable as more time is sampled.

Part 3: Nonlinear Pendulum

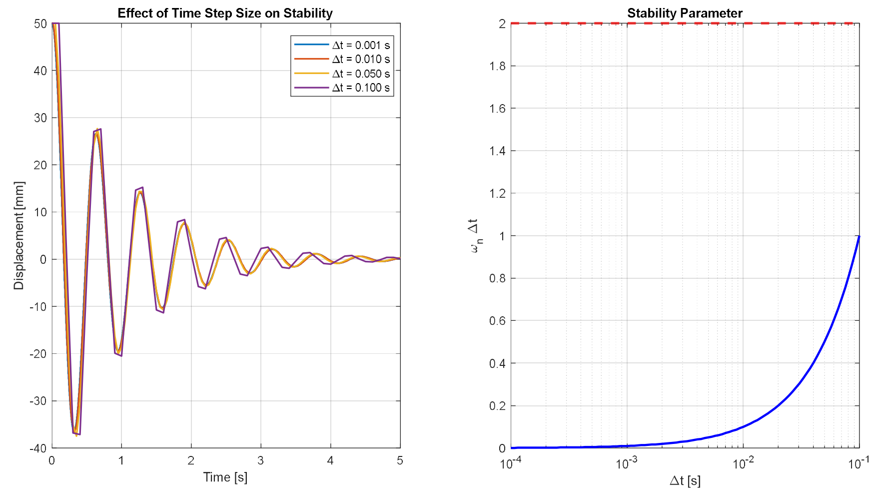
Setting a constant length of 1m we adjust the initial angles to be 15, 30, 45, and 60 degrees and initial angular velocity to be 0. The equation of a simple pendulum is $\ddot{\theta} + \frac{g}{L}\sin(\theta) = 0$ and can be linearized about $\theta = 0$ to give $\ddot{\theta} + \frac{g}{L}\theta = 0$. We again use the central difference method to numerically simulate the system.



We compare the calculated values of the linearized simulation to the nonlinear values for a large angle (60º) and a small angle (15º). We can see as time progresses the linearized model deviates from the nonlinear at a problematic rate for the large angle but for the small angle the linear model remains close enough to be a usable alternative to the computationally heavy nonlinear method. We can also test the accuracy of our model knowing that energy should be conserved, here we see the energy variety by 0.3%.

Part 4: Effects of Time Step (Stability Analysis)

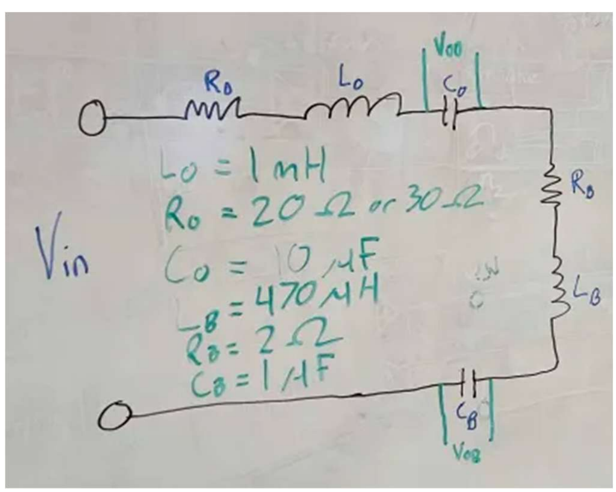
Here we tested dt values of 0.001s, 0.01s, 0.05s, and 0.1s to test the effects of time step on the stability. We use a unit step input to test the accuracy of the response.



Project

The key concepts of the project relied on the analogy created between the mass spring damper and an RCL circuit. Both acting as 2nd order systems they can be used to model one another.

While the data collected from Lab4 lead us to needing very large components the experimental side of the project was simplified to the components we had access to in the lab. This consisted of L_O = 1mH R_O = 20ohm C_O = 10uF L_B = 470mH R_B = 2ohm C_B = 1uF.



As shown in the diagram these components were then wired in series, and we measured the voltages across each of the capacitors as the outputs.

All Graphs are provided in Appendix B.

Assessment of the Results

MSD

The damping coefficient for the step response was 126Ns/m where for the frequency response was 1262 Ns/m. Looking at just the numbers the frequency response is scaled by 10 but why is this? The way we looked at the data made it so the damping coefficient relied directly on frequency and amplitude and since friction is non-linear, we had to set our amplitude different for the frequency response.

The MSD system lacks a resonance peak because the damping ratios collected through out the lab were all >1. Thus, proving the system is overdamped and therefore the magnitude plot does not have a resonance peak.

When the system is critically damped the magnitude plot would still not show a resonance peak. It would look like an underdamped system but shot up faster and level out with not oscillation. Whereas if the damping ration was <0.707 a resonance peak would appear right around natural frequency.

Friction is a big source of error. The standard MSD differential equations is linear, but we have a strong non-linear force due to friction. This varies many values needed in this lab make the data less consistent and accurate. Another error is reading the phase shifts correctly. Measuring phase shift by eye was difficult and lead to some estimations cause another source of error.

BEAM

Model Validation

As there was no optical sensor in the experiment comparing values is unreliable, however we can still explain the factors that may cause discrepancies between experimental and theoretical results. One key factor is that friction behaves as a non-linear damper so the linear viscus model does not completely match up with the real system. Sensor noise always plays a role, however if it is mostly random it can be reduced through averages.

System Parameters

For the beam exact values/graphs for frequency and motion were unobtainable, however the equations one would use given access to this data would be:

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$$

Where zeta can be calculated through either percent overshoot:

$$\zeta = -\frac{\ln\left(\frac{PO}{100}\right)}{\pi^2 + \ln^2\left(\frac{PO}{100}\right)}$$

Or through logarithmic decrement:

$$\delta = \ln\left(\frac{x_1}{x_2}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

Expected values from Damping Properties of Materials [2], are metals in elastic range typically suggest damping ratios below 0.01 and continuous metal structures show damping ratios between 0.02-0.04.

Friction Effects

OSCAR increases the apparent damping ratio beyond the true mechanical damping of the beam. The observed damping from the experiments will encapsulate the friction combined with the beams damping which causes the beam be viewed with larger damping then the material normally has. Example most metal beam structures have a damping ratio less than 0.05.

Poles and System Behaviour

The characteristic equation in the S-plane is:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

With pole locations found at:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$$

For an underdamped system like our beam was, ($\zeta < 1$)

- The real part controls the decay rate and settling time, with a more negative value making a faster settling time.
- Imaginary part controls the oscillation frequency, larger causing faster oscillations
- Distance from origin is the overall system speed
- Angle from negative real axis relates to damping ratio, smaller angle less damping.

Zero Location

The transfer function has zeros at:

$$s = -\frac{\omega_n^2}{2\zeta\omega_n} = -\frac{\omega_n}{2\zeta}$$

If the Zero location is further left (more negative higher Zeta)

- Less influence on transient response, lower overshoot, response closer to standard second order system.

Further Right (less negative, closer to imaginary axis)

- Stronger influence on response, can cause undershoot if zero is too close to poles

At Origin (s=0)

- Eliminates steady state response to step input, system acts as high-pass filter.

Connection to Frequency Response

- At low frequencies the beam follows Oscars motion meaning they are in phase.
- At resonance frequency the beam has a phase difference of 90°, maximum beam amplitude and large oscillations. This kind of frequency is catastrophic for destructible structures.
- At high frequencies the beam has a phase difference of 180°, causing the beam to move opposite OSCAR. Beam amplitude is very small while the base moves.

Extension Question

If the beam were to be taller the natural frequency would be impacted following this equation:

$$\omega_n = \sqrt{\frac{k}{m}}\ \alpha\ \sqrt{\frac{EI}{L^3 m_{tip}}}$$

Taller beams have larger L, causing the natural frequency to decrease.
More flexible beams have lower E or I cause natural frequency to decrease.
The damping ratio however is mostly a property of the material, possible that it slightly increases with flexibility however expected to remain relatively constant.

Real world analogies:

- Tall buildings behave very similarly to our beam for small oscillations however buildings tend to collapse when oscillations grow too large.

Any real-world scenario revolving around a tall structure requires dealing with base excitations ensuring that the structure never passes safe limits and all vibrations are dampened.

Conclusion

Overall, this lab helps us understand how a MSD system behaves and how a second order equation could describe it. From the step response test that we obtained with OSCAR, we found that the system was overdamped and there no resonance in the magnitude plot. This is due too the friction that effect the motion, this is also the reason why we did not see a sharp resonance peak in the frequency response. When we estimated the m, c, and k using step response, frequency response, and numerical method all of them gave slightly different results, this showed how the friction in OSCAR is nonlinear.

From the pole analysis and s plane, we found that the real and imaginary parts determine the decay rate and oscillation. The friction in OSCAR increases the damping ratio compared to the metal beam would normally have. The zero-location helped explain changes in overshoot and how the beam response shift depending on where the zero sits relative to the poles. The frequency response results that we found matched the expected trends, the beam followed OSCAR at low frequencies, showed large oscillation and a 90-degree phase shift near resonance.

Converting the MSD model into the RLC circuit showed how mechanical system can represented using electrical components. The Lab 5 simulations, including the forced vibration, nonlinear pendulum, and time step showed how changing damping and time step affects the amplitude, phase, and numerical stability. This lab demonstrated that second order models are very useful for predicting and understanding the behaviour of a systems. Real systems always include friction, noise and nonlinear affects, so the experimental and theoretical values will never match perfectly. Another limitation of the OSCAR system in its current state is that the system lacks the necessary electronics to measure the beam.

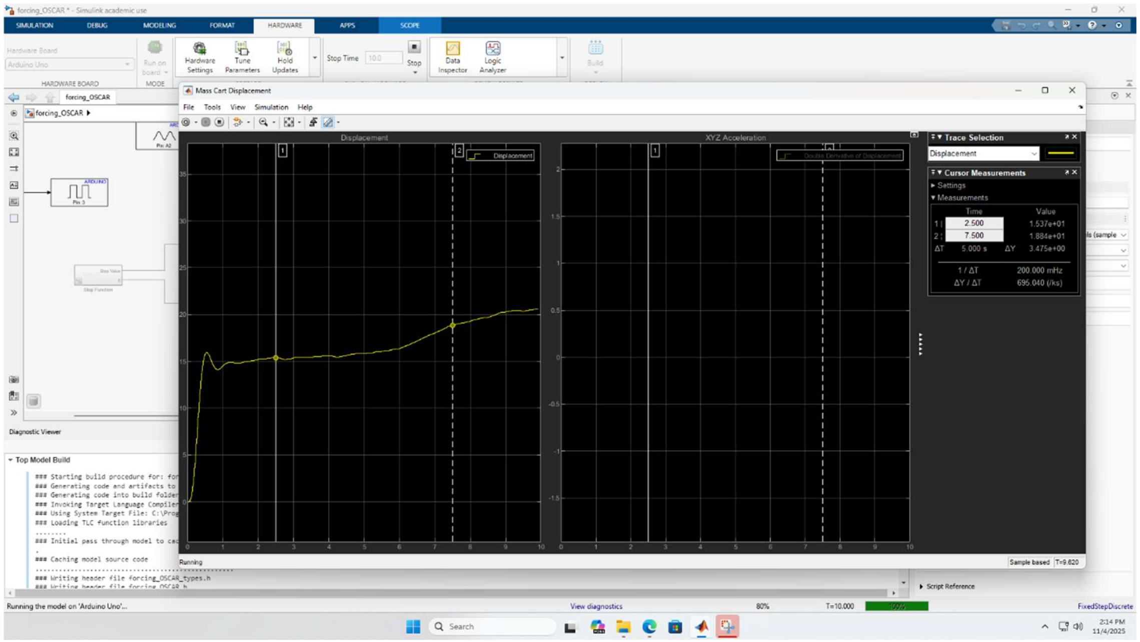
References

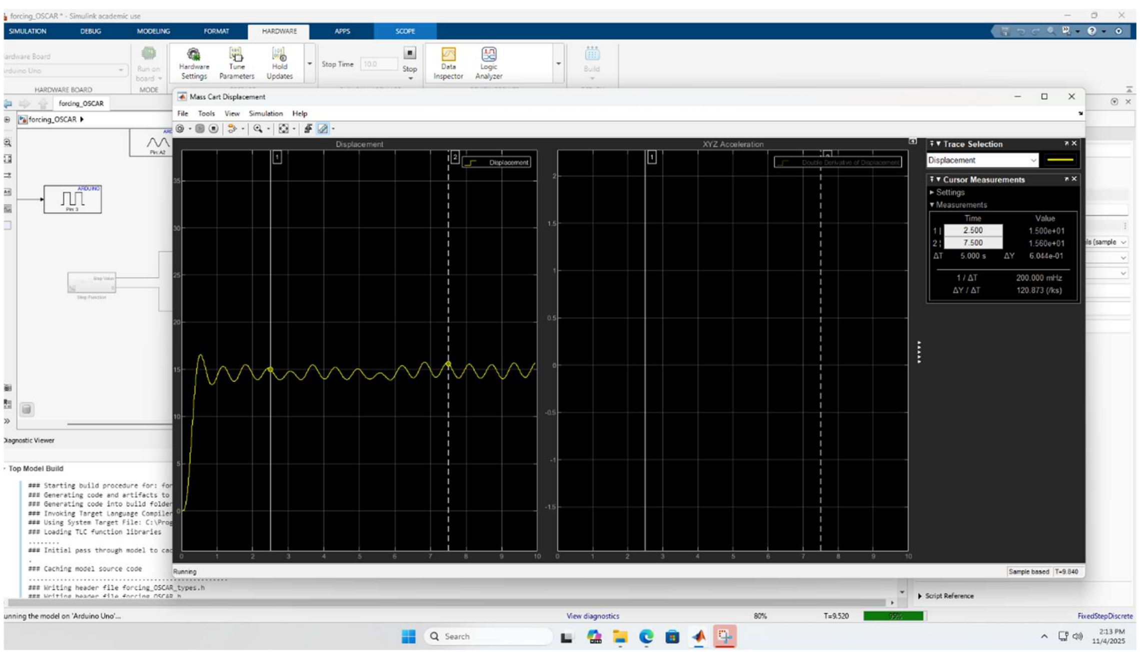
[1] F. Golnaraghi, " Lab 4: Forced Response Analysis of a Mass-Spring-Damper," 2025.

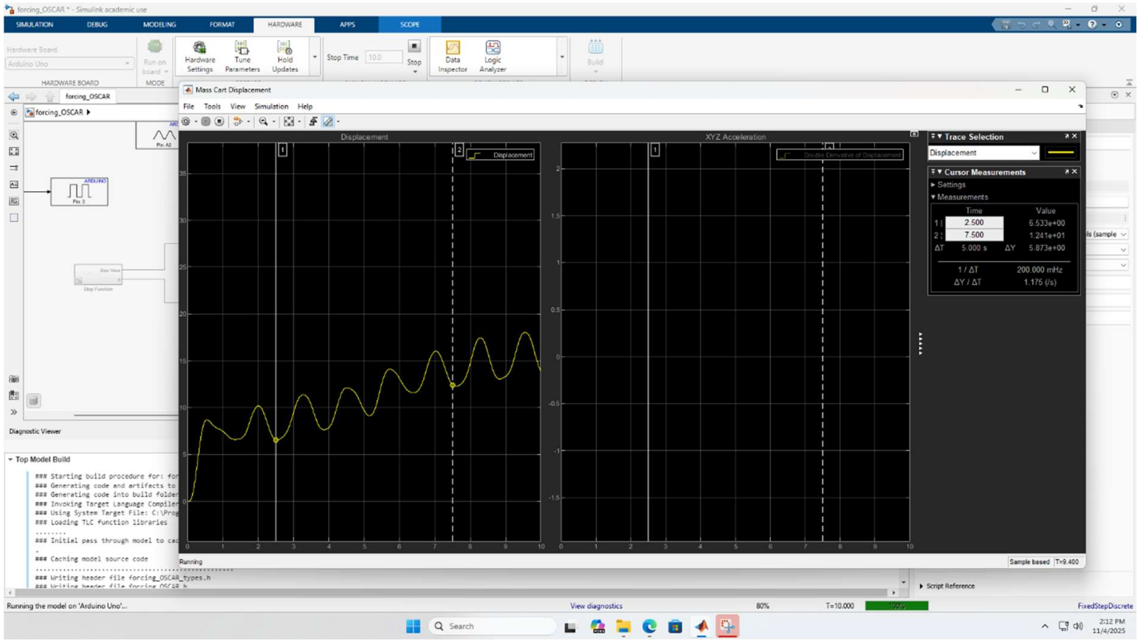
[2] Irvine, T. "Damping Properties of Materials," Revision D. Available at: "http://www.vibrationdata.com/tutorials_alt/damping.pdf" 2025.

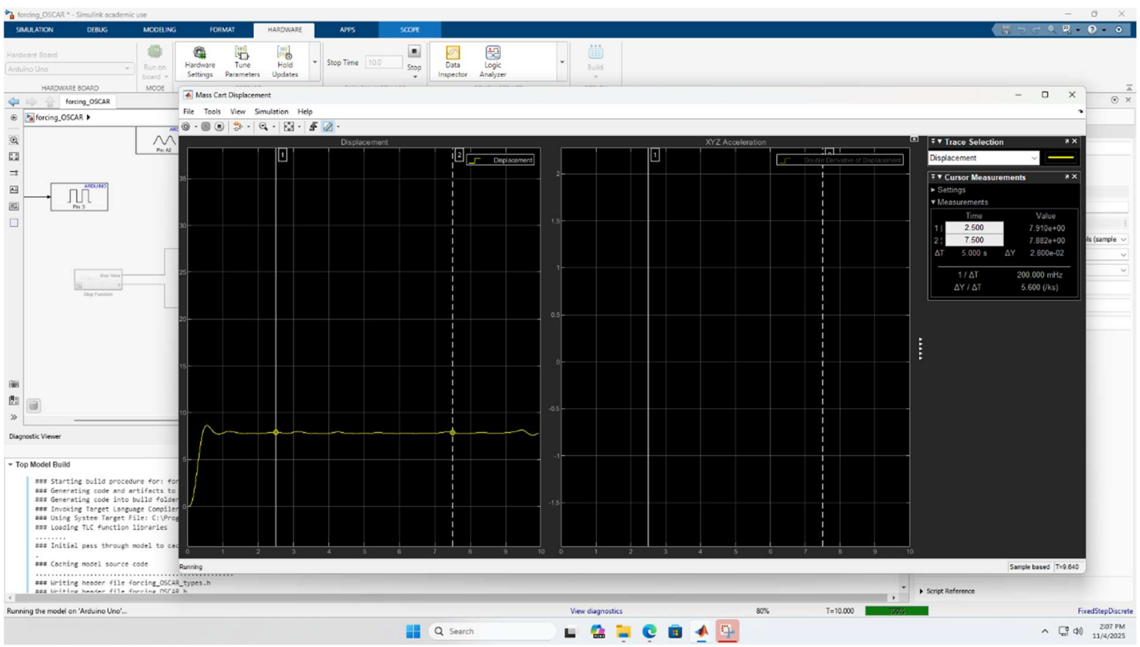
Appendix

Appendix A






Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements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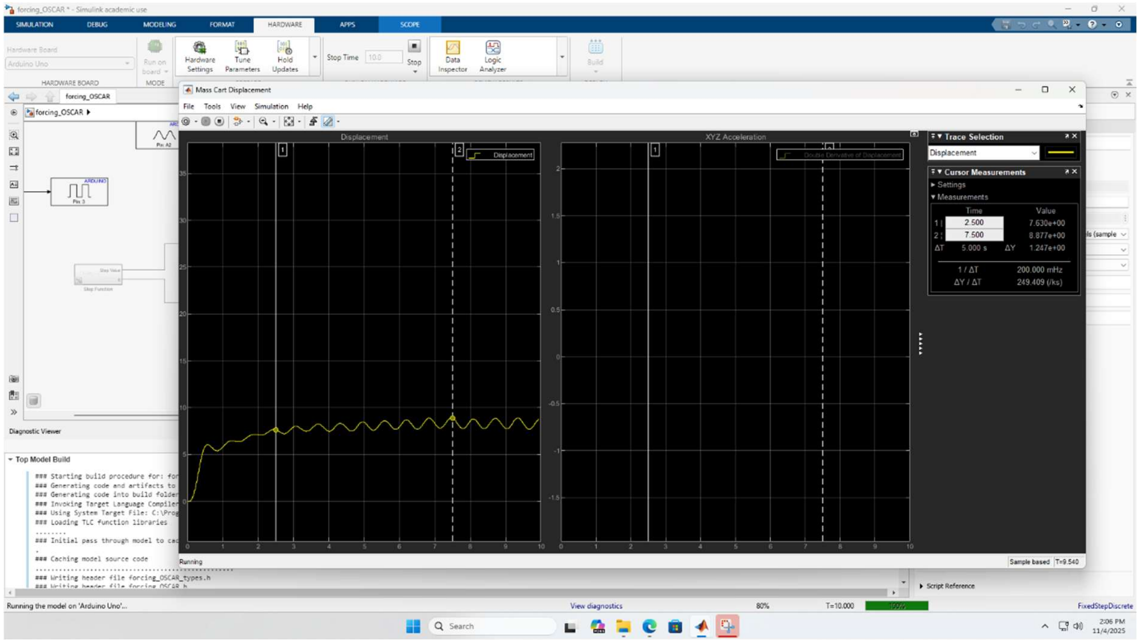
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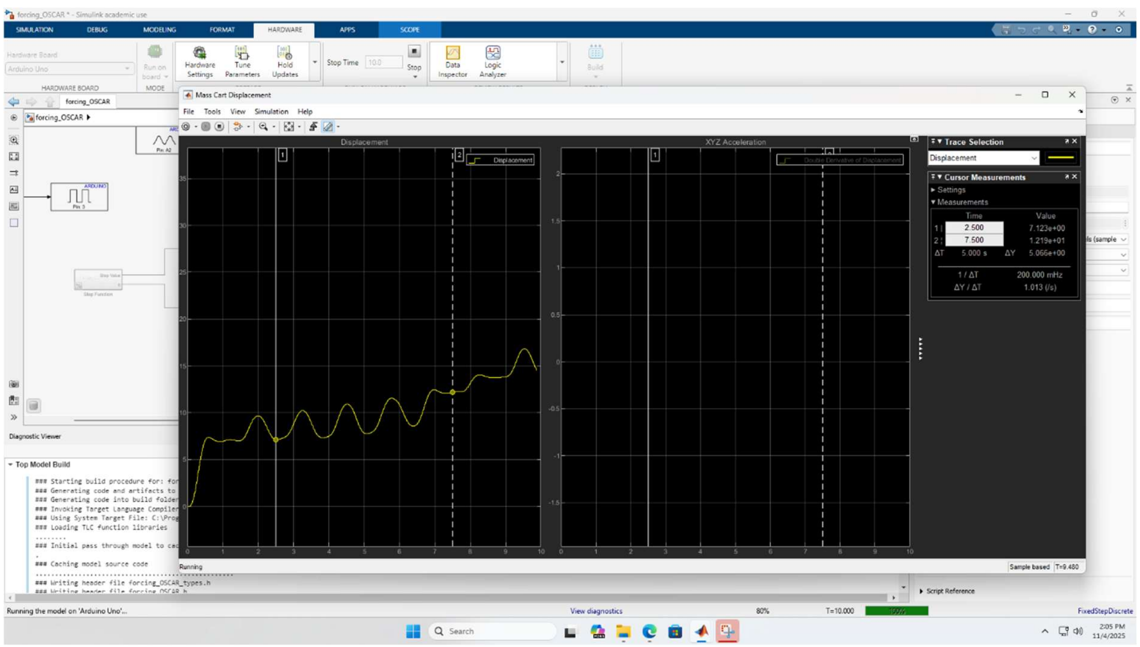
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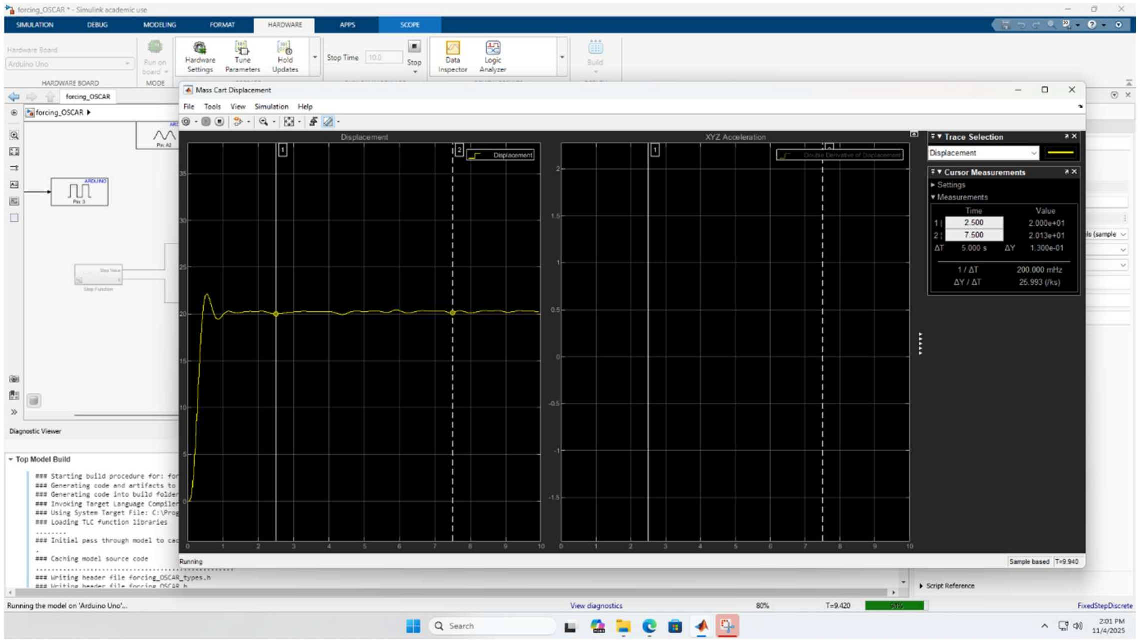
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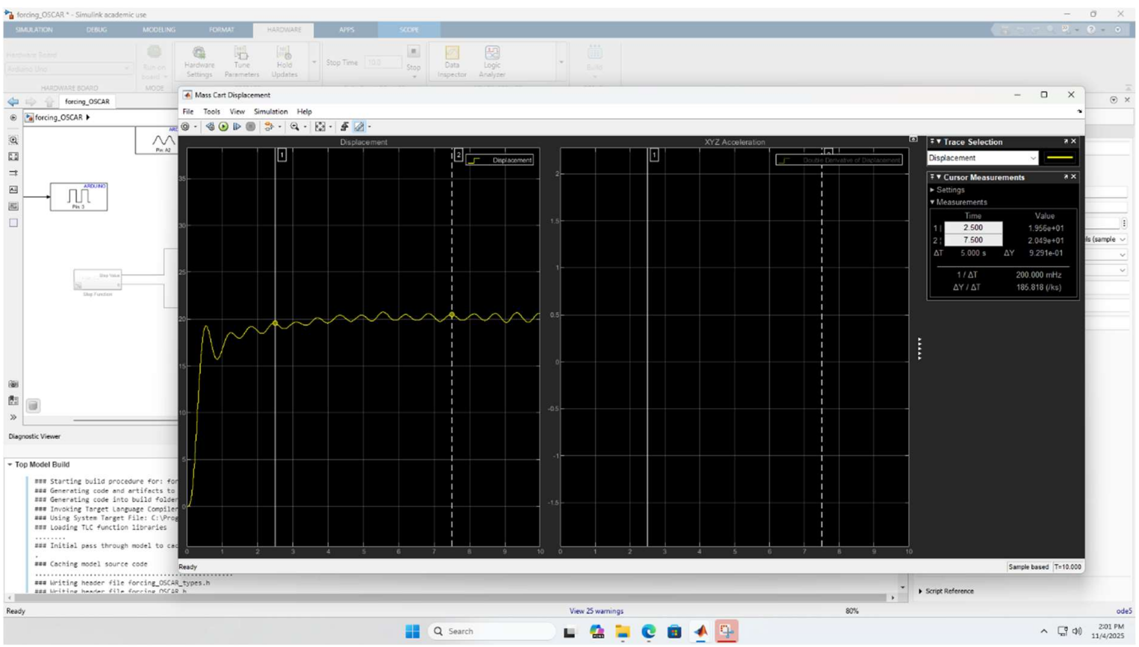
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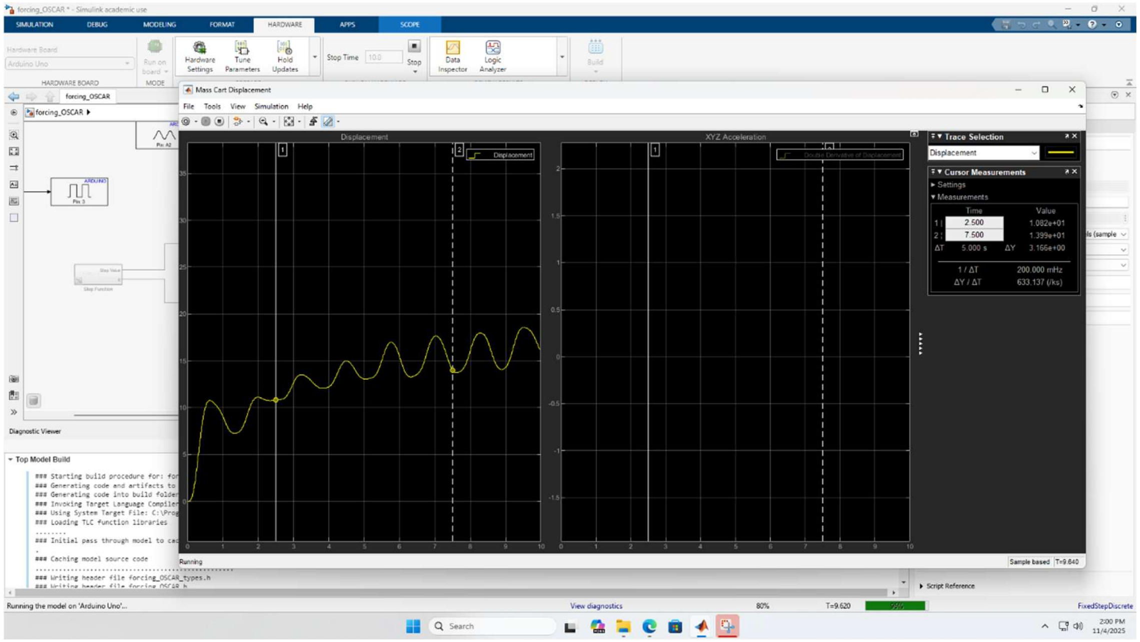
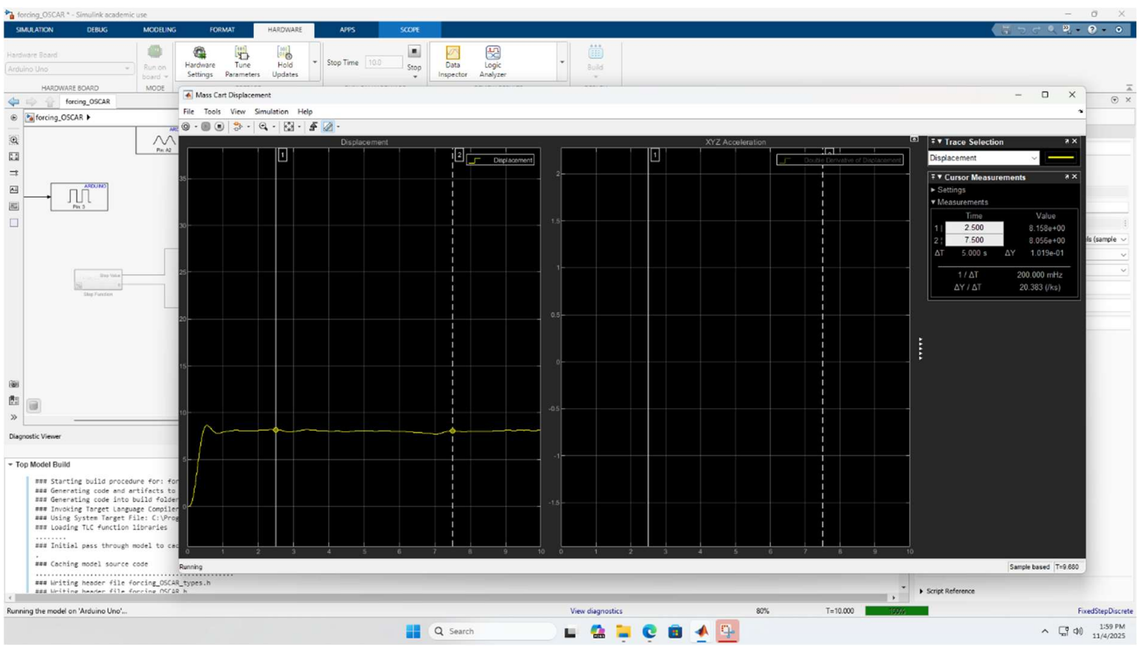
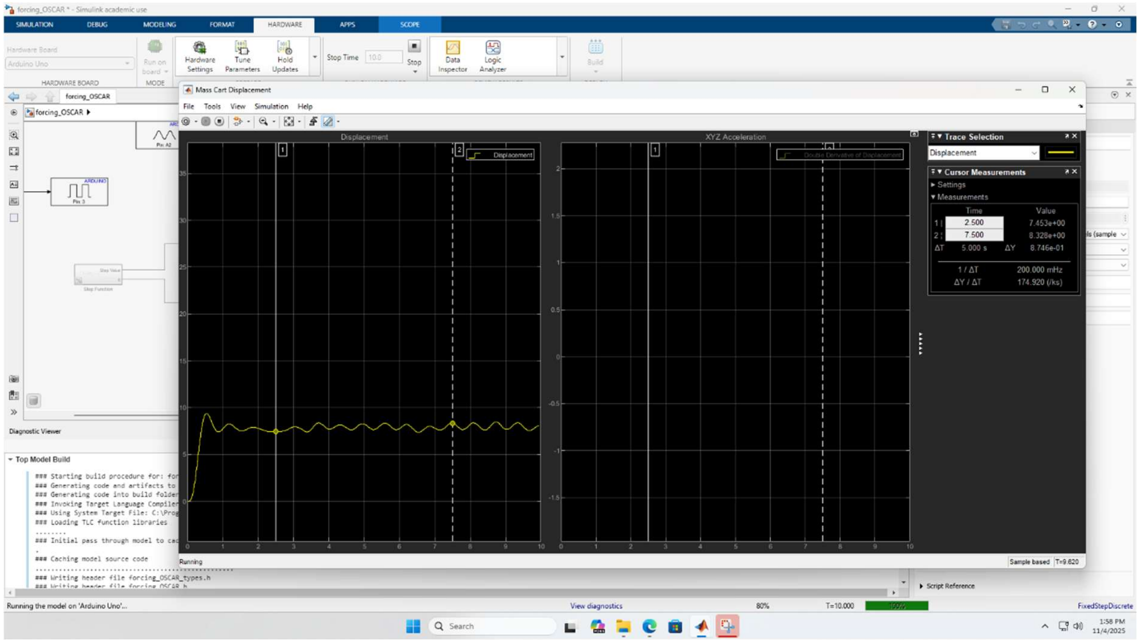
XYZ Acceleration
Cursor Measurements



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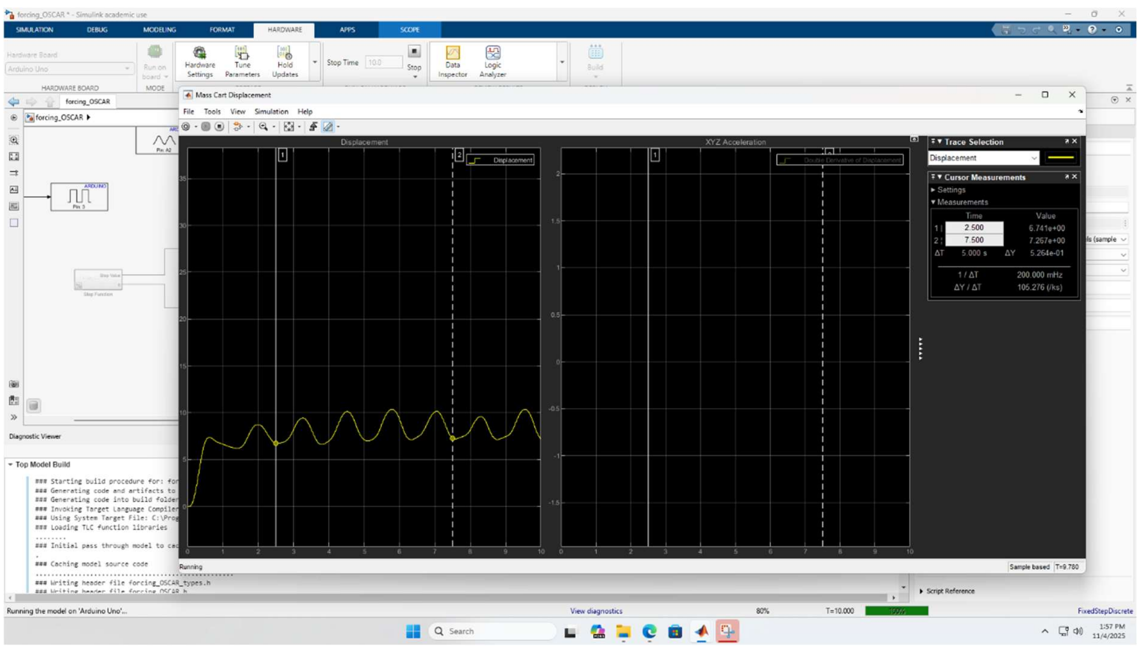
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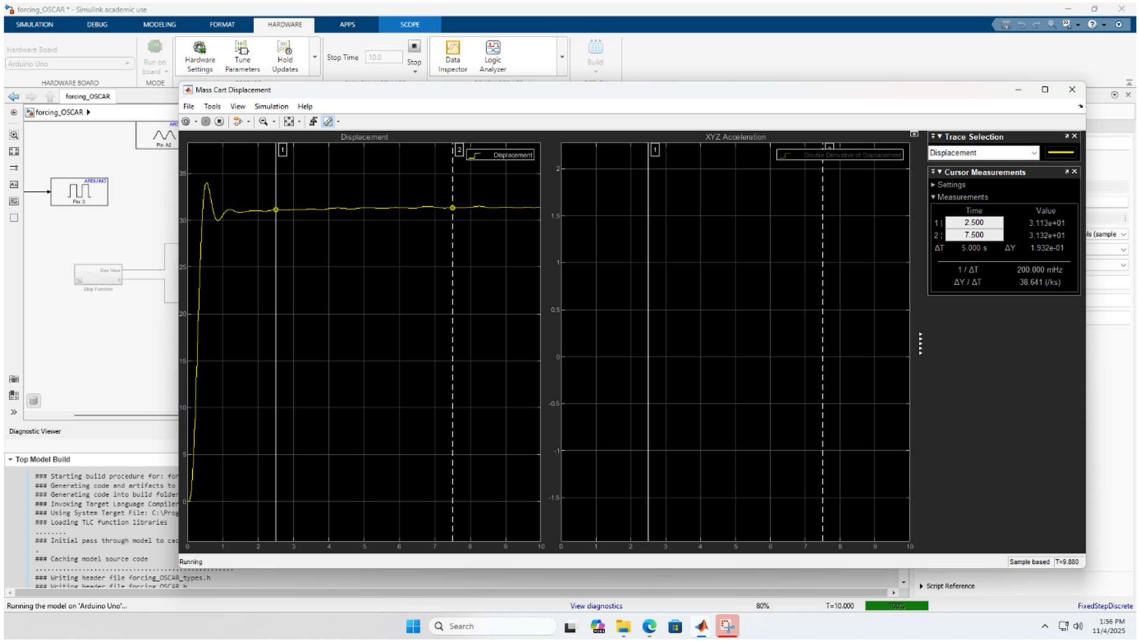
Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
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7.500
3.113e+01
3.132e+01
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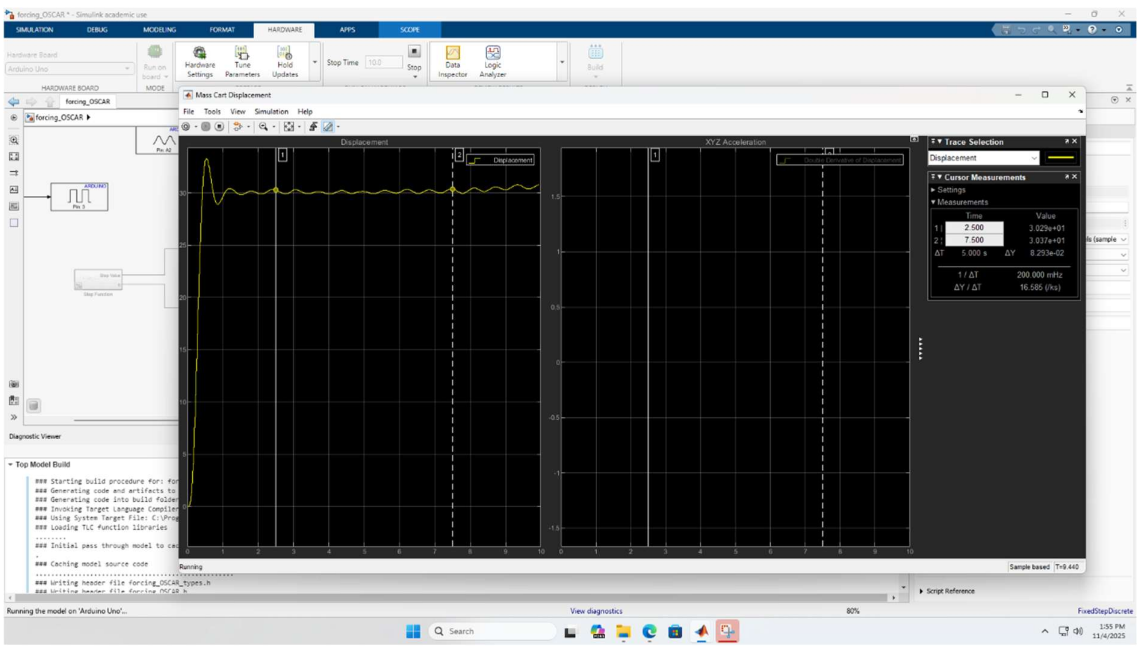
1.932e-01
200.000 mHz
38.641 (/ks)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Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
Time
Value
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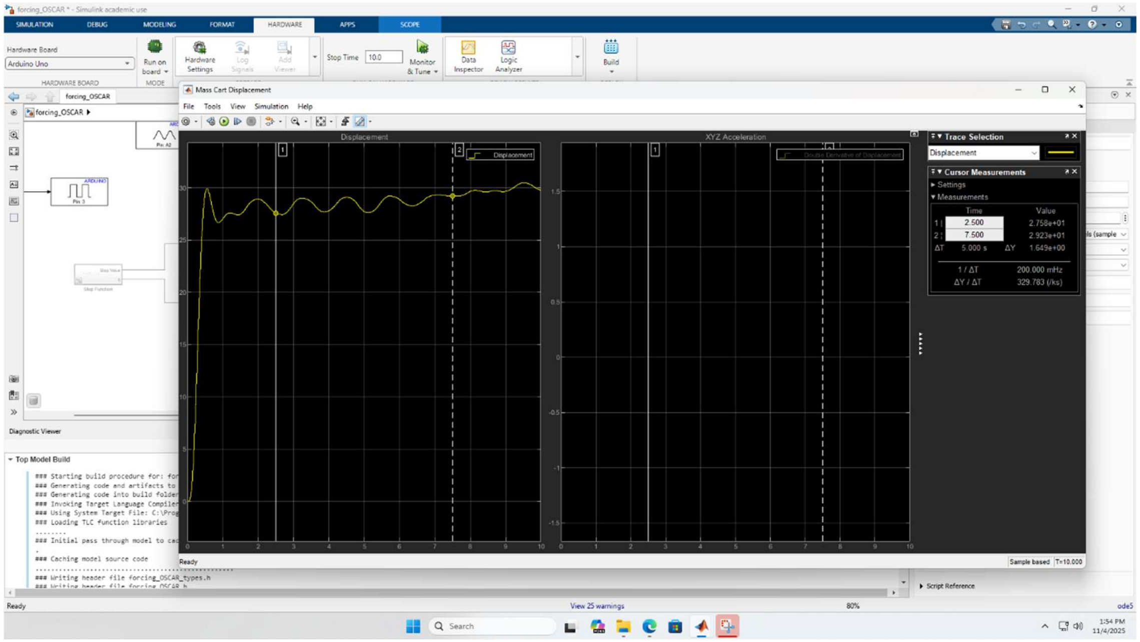
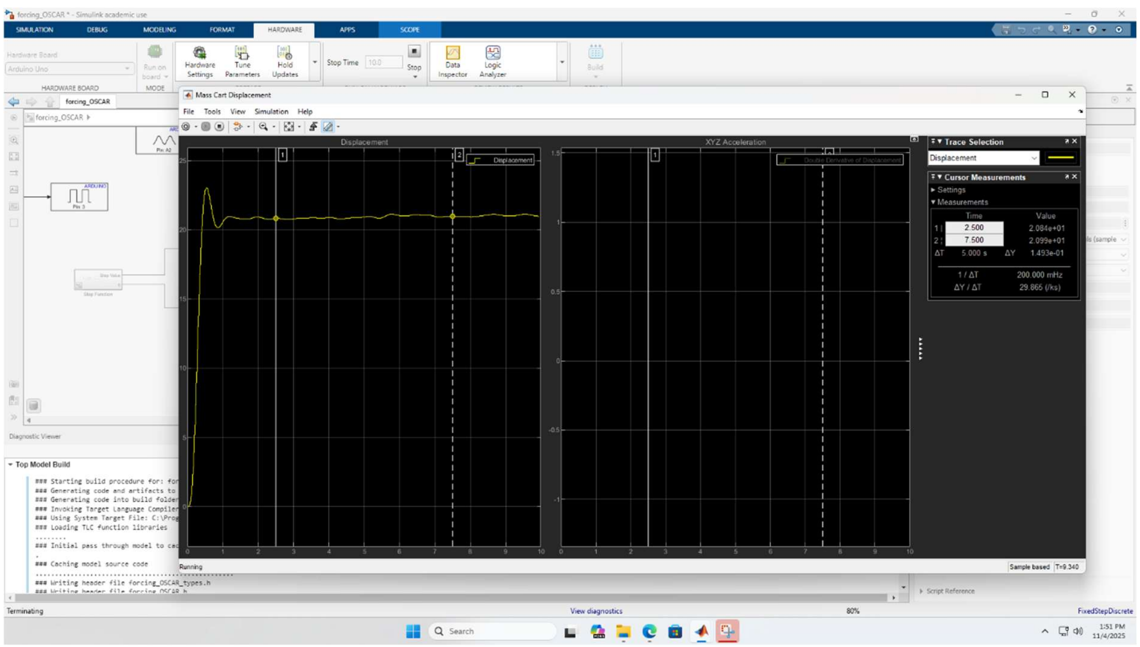
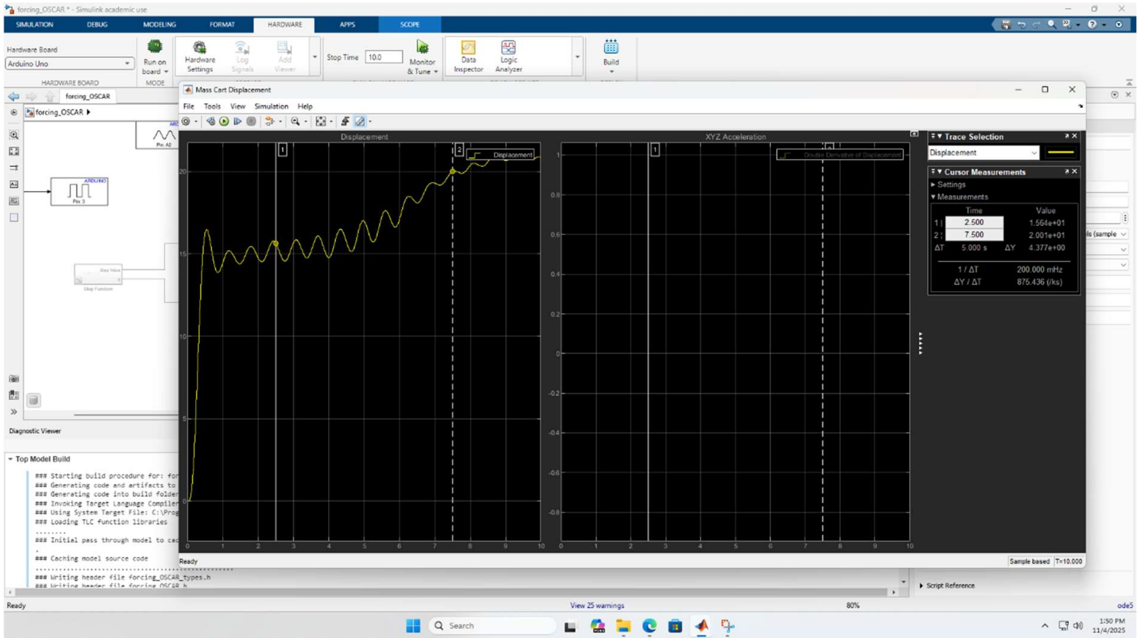
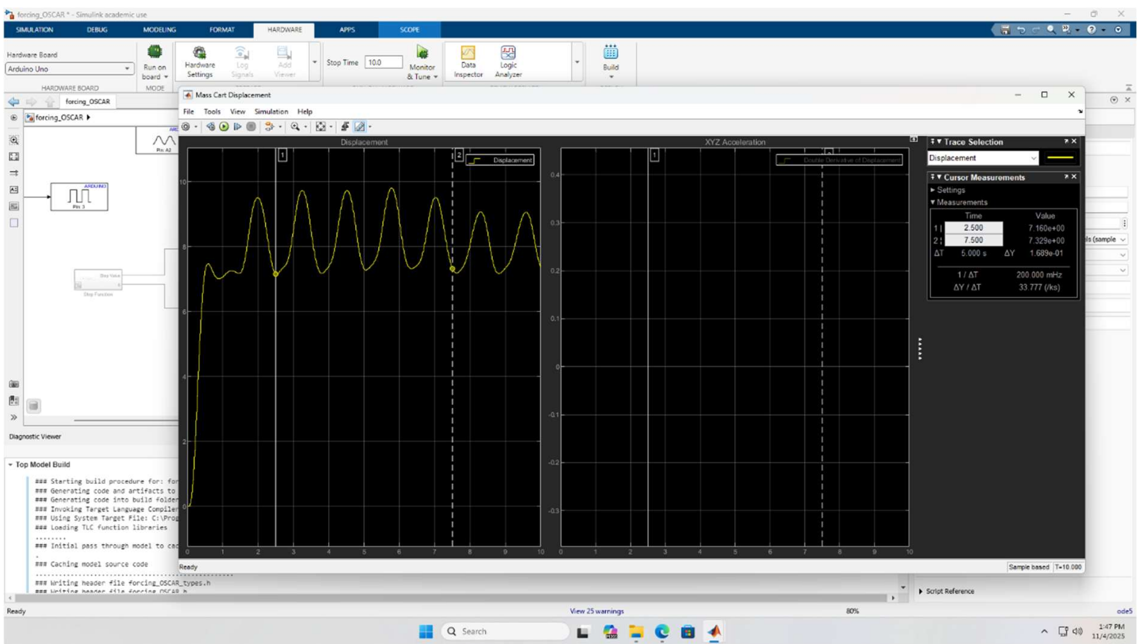
7.500
3.029e+01
3.037e+01
5.000 s
8.293e-02
200.000 mHz
16.586 (/ks)


Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
Time
Value
2.500
7.500
5.000 s
200.000 mHz
Ready


Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
Time
Value
2.500
7.500
5.000 s
200.000 mHz
Running

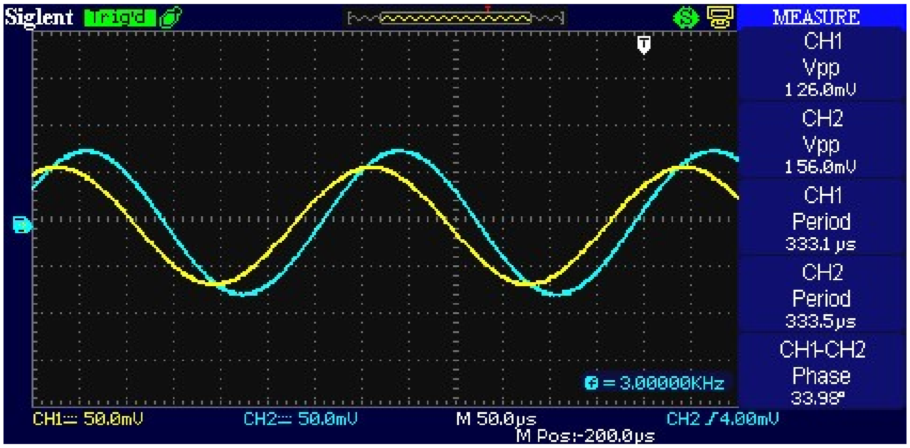


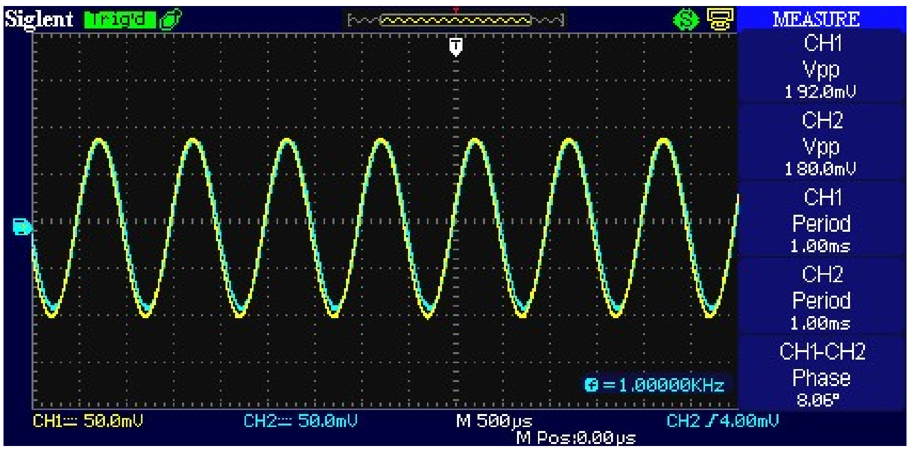
forcing_OSCAR * - Simulink academic use
Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
Time
Value
2.500
7.500
1.564e+01
2.001e+01
5.000 s
4.377e+00
200.000 mHz
875.436 (/ks)

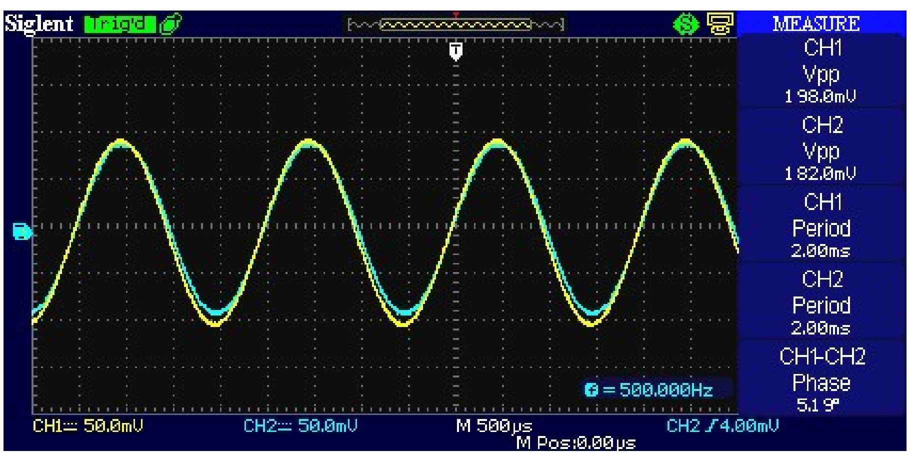


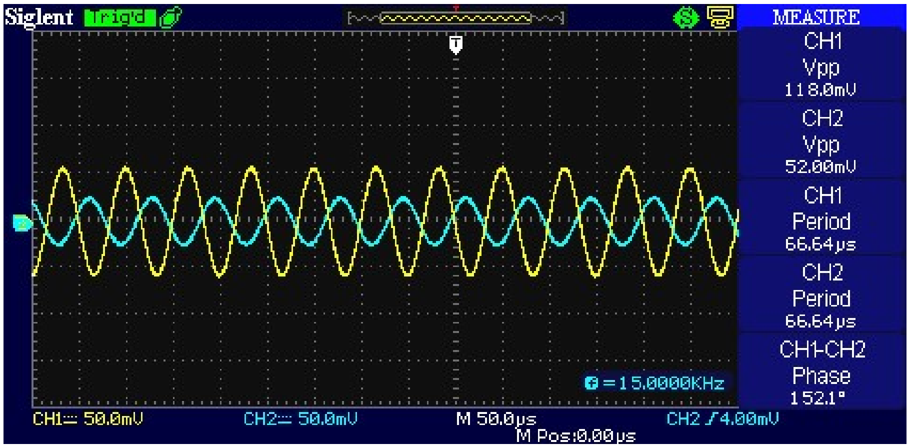
forcing_OSCAR * - Simulink academic use
Mass Cart Displacement
Displacement
XYZ Acceleration
Trace Selection
Cursor Measurements
Time
Value
2.500
7.500
7.160e+00
7.329e+00
5.000 s
1.689e-01
200.000 mHz
33.777 (/ks)

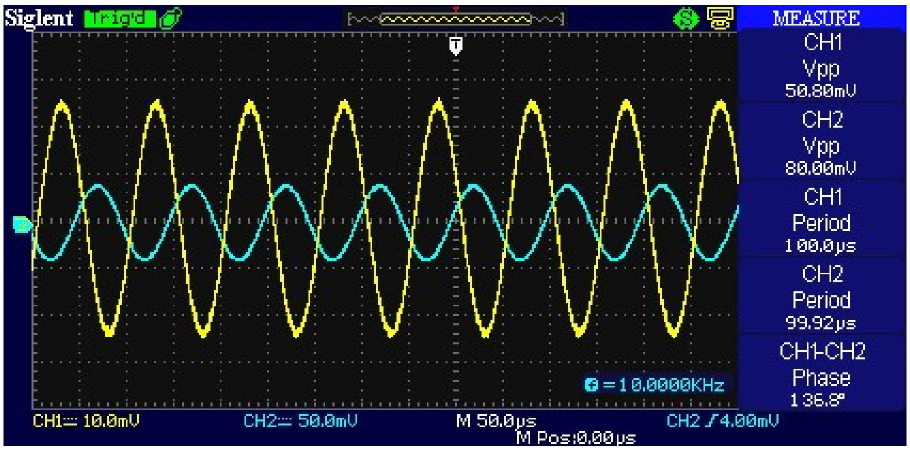
Appendix B

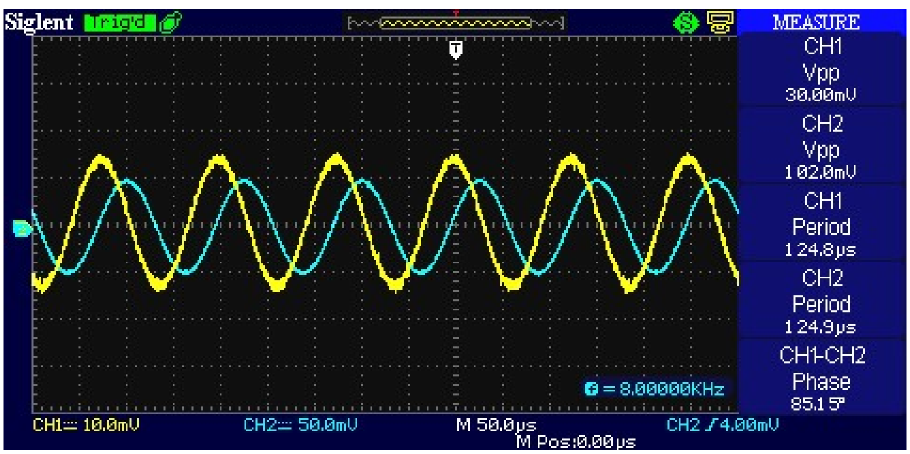

Siglent
Trig'd
MEASURE
CH1
Vpp
126.0mV
CH2
Vpp
156.0mV
CH1
Period
333.1μs
CH2
Period
333.5μs
CH1-CH2
Phase
33.98°
= 3.00000KHz
CH1 ⎓ 50.0mV
CH2 ⎓ 50.0mV
M 50.0μs
M Pos:-200.0μs
CH2 ⨍ 4.00mV

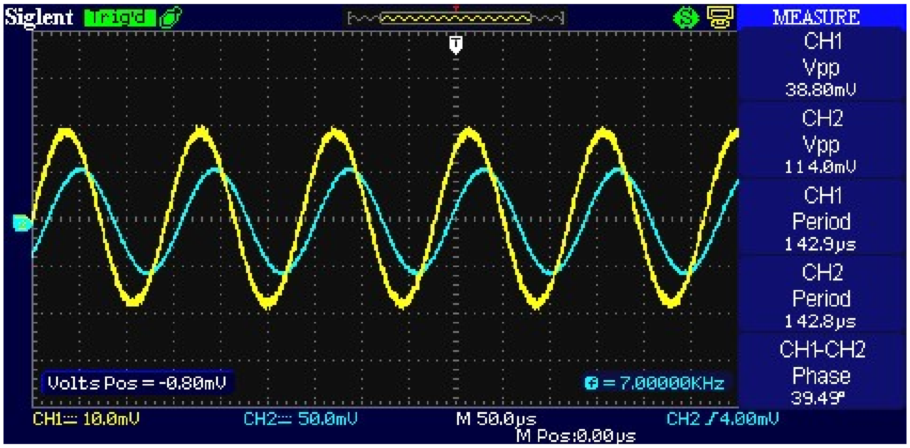

Siglent
Trig'd
MEASURE
CH1
Vpp
192.0mV
CH2
Vpp
180.0mV
CH1
Period
1.00ms
CH2
Period
1.00ms
CH1-CH2
Phase
8.06°
= 1.00000KHz
CH1 ⎓ 50.0mV
CH2 ⎓ 50.0mV
M 500μs
M Pos:0.00μs
CH2 ⨍ 4.00mV

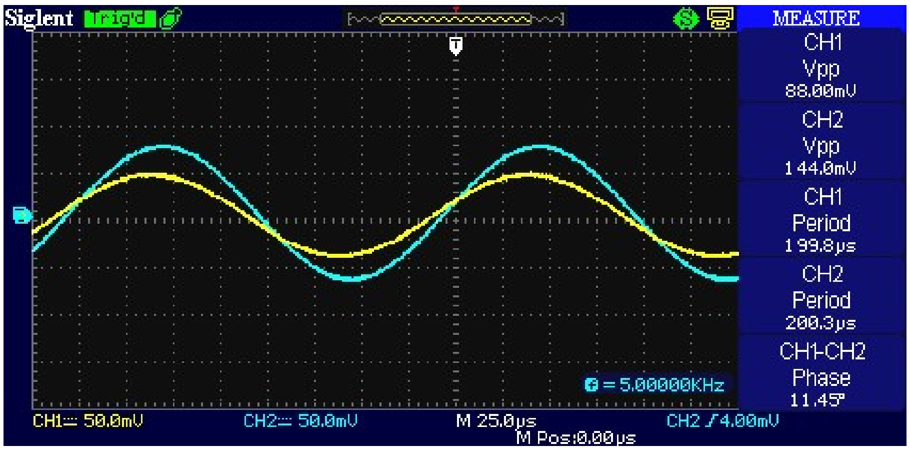

Siglent
Trig'd
MEASURE
CH1
Vpp
198.0mV
CH2
Vpp
182.0mV
CH1
Period
2.00ms
CH2
Period
2.00ms
CH1-CH2
Phase
5.19°
= 500.000Hz
CH1 ⎓ 50.0mV
CH2 ⎓ 50.0mV
M 500μs
M Pos:0.00μs
CH2 ⨍ 4.00mV

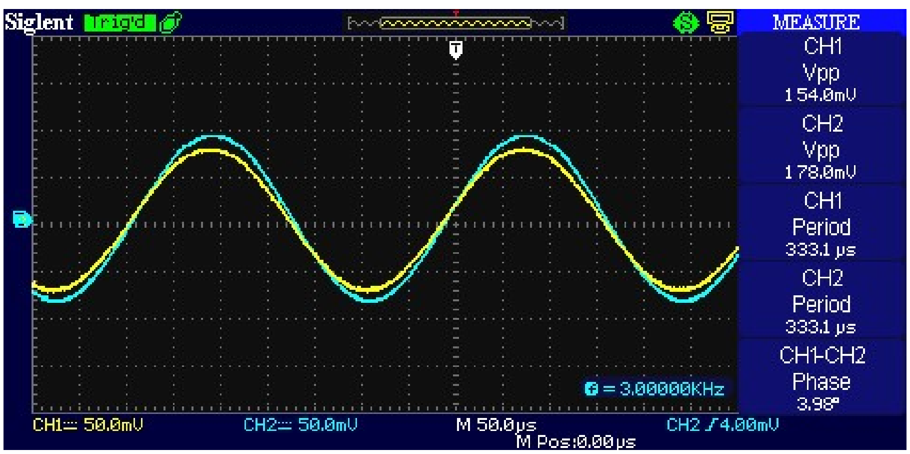

Siglent
Trig'd
MEASURE
CH1
Vpp
118.0mV
CH2
Vpp
52.00mV
CH1
Period
66.64µs
CH2
Period
66.64µs
CH1-CH2
Phase
152.1°
=15.0000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0µs
M Pos:0.00µs
CH2 4.00mV

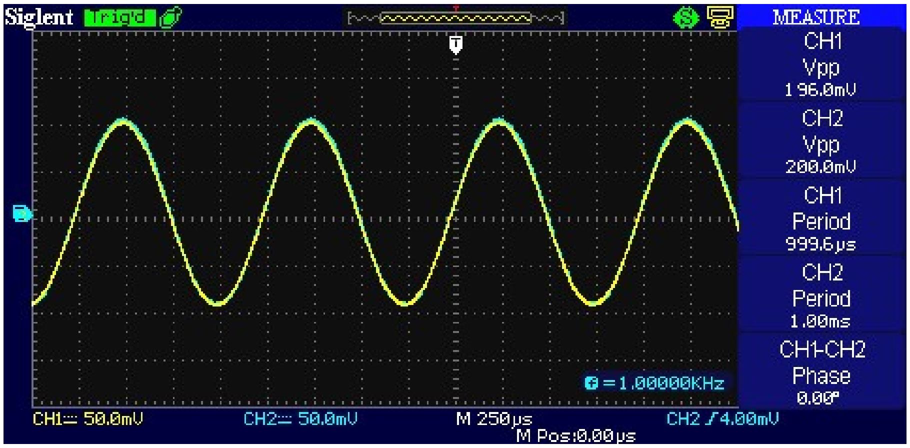

Siglent
Trig'd
MEASURE
CH1
Vpp
50.80mV
CH2
Vpp
80.00mV
CH1
Period
100.0µs
CH2
Period
99.92µs
CH1-CH2
Phase
136.8°
=10.0000KHz
CH1 10.0mV
CH2 50.0mV
M 50.0µs
M Pos:0.00µs
CH2 4.00mV

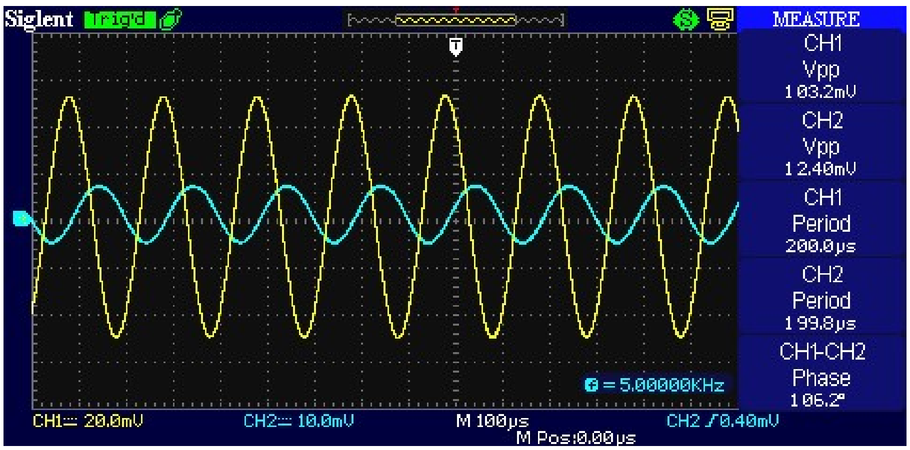

Siglent
Trig'd
MEASURE
CH1
Vpp
30.00mV
CH2
Vpp
102.0mV
CH1
Period
124.8µs
CH2
Period
124.9µs
CH1-CH2
Phase
85.15°
=8.00000KHz
CH1 10.0mV
CH2 50.0mV
M 50.0µs
M Pos:0.00µs
CH2 4.00mV

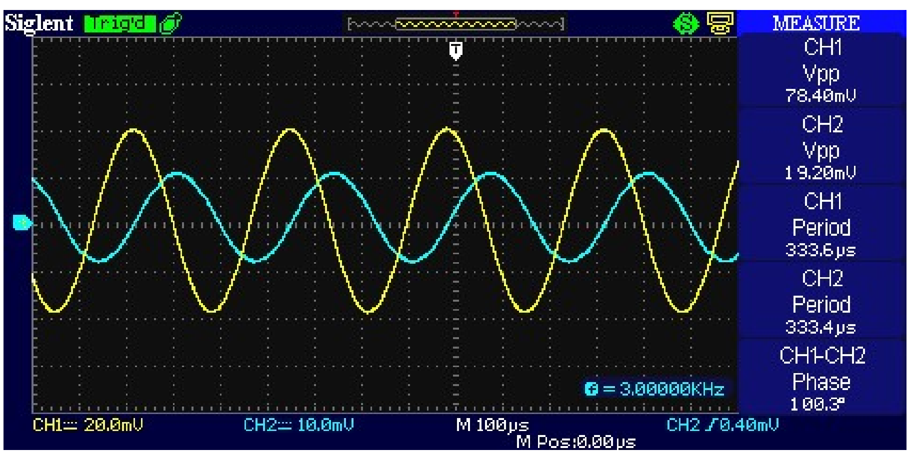

Siglent Trig'd
MEASURE
CH1 Vpp 38.80mV
CH2 Vpp 114.0mV
CH1 Period 142.9µs
CH2 Period 142.8µs
CH1-CH2 Phase 39.49°
Volts Pos = -0.80mV
= 7.00000KHz
CH1 10.0mV
CH2 50.0mV
M 50.0µs
M Pos:0.00µs
CH2 4.00mV

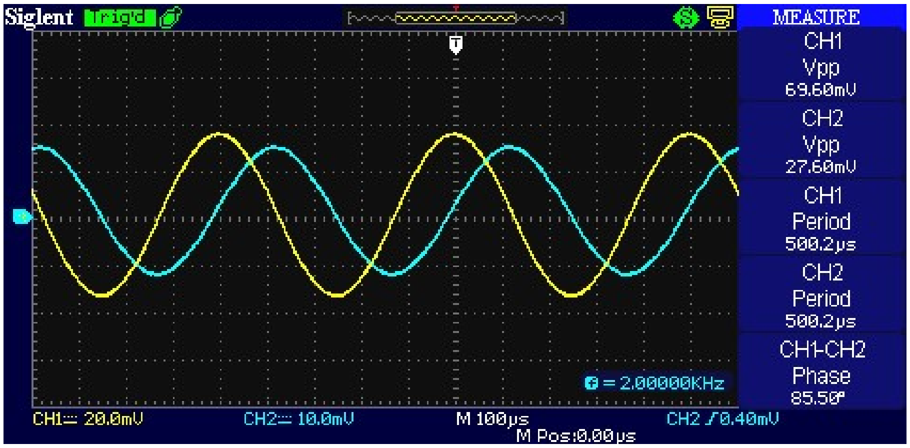

Siglent Trig'd
MEASURE
CH1 Vpp 88.00mV
CH2 Vpp 144.0mV
CH1 Period 199.8µs
CH2 Period 200.3µs
CH1-CH2 Phase 11.45°
= 5.00000KHz
CH1 50.0mV
CH2 50.0mV
M 25.0µs
M Pos:0.00µs
CH2 4.00mV

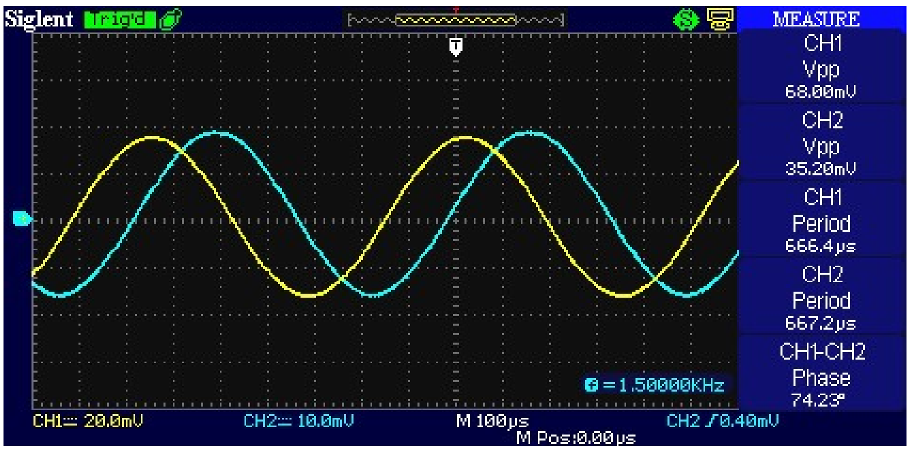

Siglent Trig'd
MEASURE
CH1 Vpp 154.0mV
CH2 Vpp 178.0mV
CH1 Period 333.1µs
CH2 Period 333.1µs
CH1-CH2 Phase 3.98°
= 3.00000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0µs
M Pos:0.00µs
CH2 4.00mV

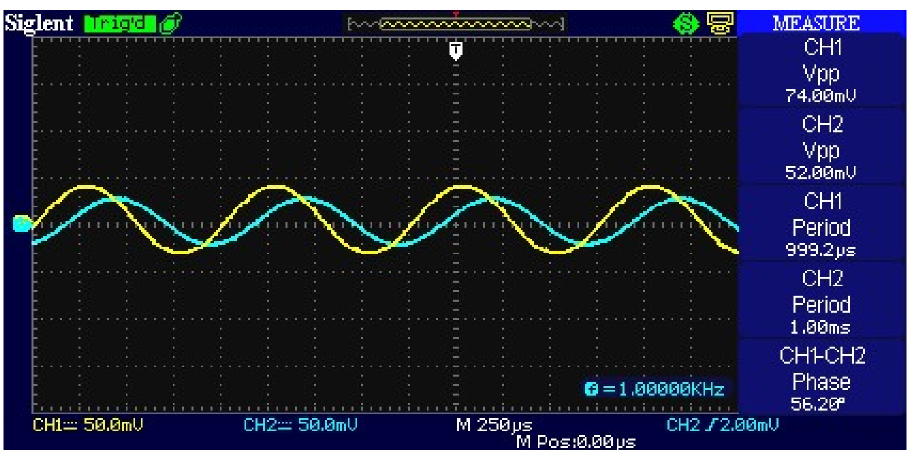

Siglent
Trig'd
MEASURE
CH1
Vpp
196.0mV
CH2
Vpp
200.0mV
CH1
Period
999.6µs
CH2
Period
1.00ms
CH1-CH2
Phase
0.00°
=1.00000KHz
CH1 50.0mV
CH2 50.0mV
M 250µs
M Pos:0.00µs
CH2 4.00mV

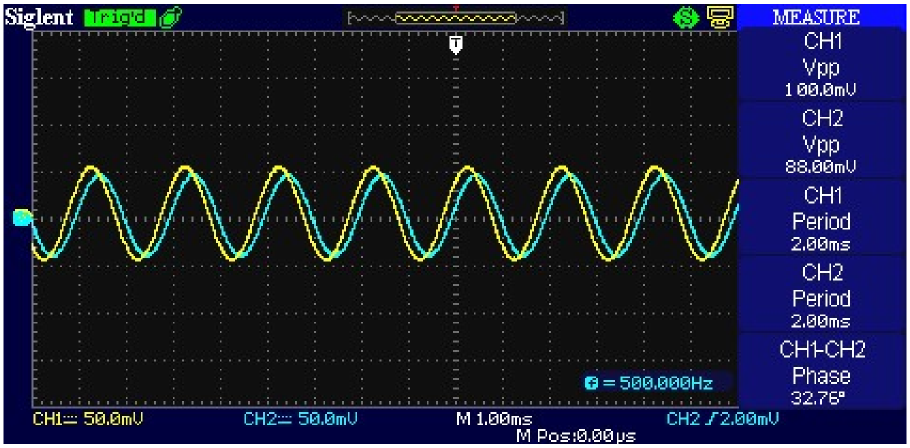

Siglent
Trig'd
MEASURE
CH1
Vpp
103.2mV
CH2
Vpp
12.40mV
CH1
Period
200.0µs
CH2
Period
199.8µs
CH1-CH2
Phase
106.2°
=5.00000KHz
CH1 20.0mV
CH2 10.0mV
M 100µs
M Pos:0.00µs
CH2 0.40mV

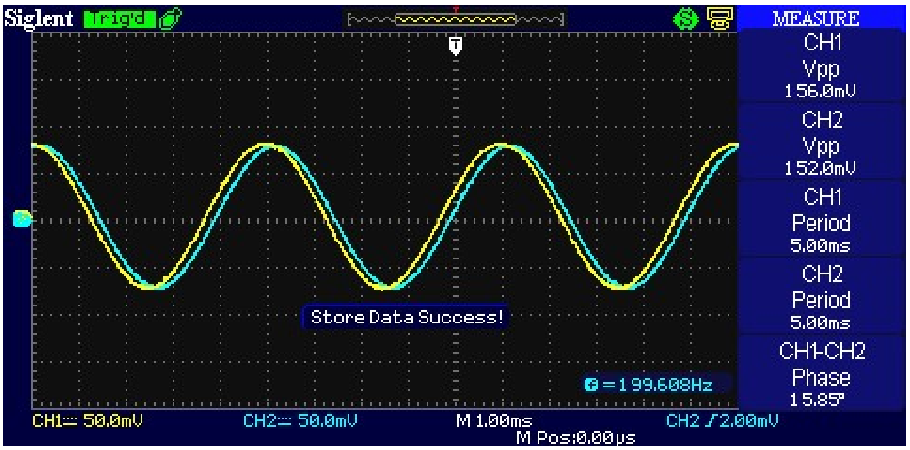

Siglent
Trig'd
MEASURE
CH1
Vpp
78.40mV
CH2
Vpp
19.20mV
CH1
Period
333.6µs
CH2
Period
333.4µs
CH1-CH2
Phase
100.3°
=3.00000KHz
CH1 20.0mV
CH2 10.0mV
M 100µs
M Pos:0.00µs
CH2 0.40mV

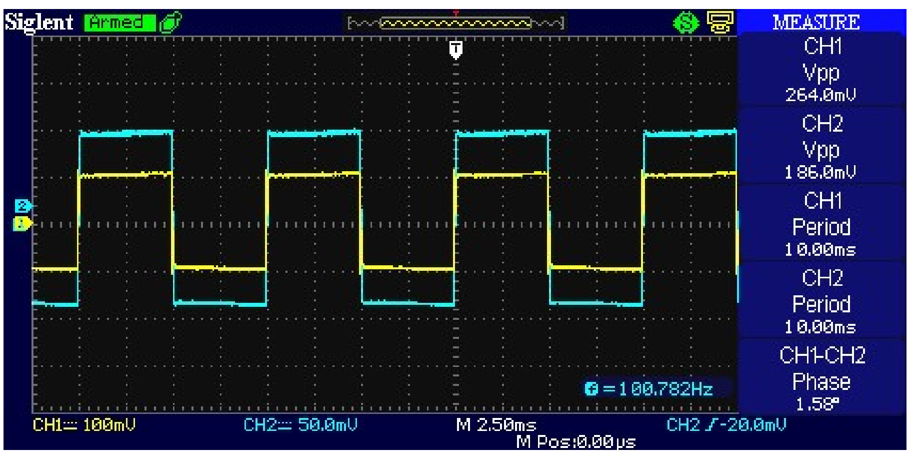

Siglent
Trig'd
MEASURE
CH1
Vpp
69.60mV
CH2
Vpp
27.60mV
CH1
Period
500.2μs
CH2
Period
500.2μs
CH1-CH2
Phase
85.50°
=2.00000KHz
CH1⎓ 20.0mV
CH2⎓ 10.0mV
M 100μs
M Pos:0.00μs
CH2 ∫0.40mV

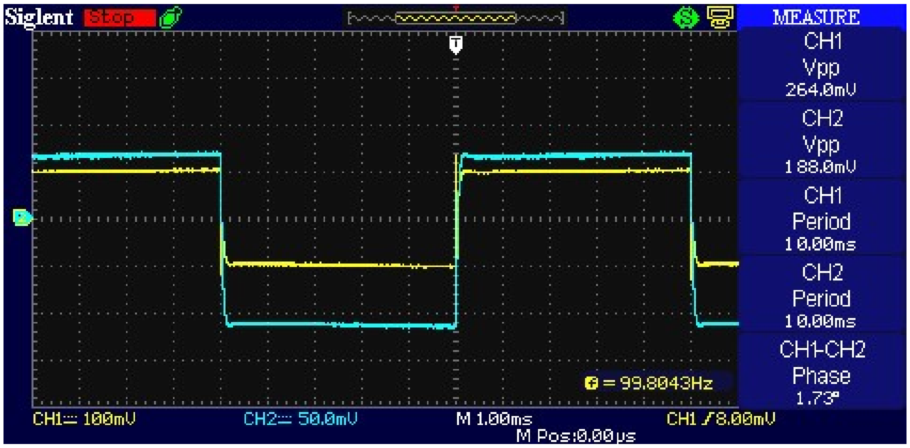

Siglent
Trig'd
MEASURE
CH1
Vpp
68.00mV
CH2
Vpp
35.20mV
CH1
Period
666.4μs
CH2
Period
667.2μs
CH1-CH2
Phase
74.23°
=1.50000KHz
CH1⎓ 20.0mV
CH2⎓ 10.0mV
M 100μs
M Pos:0.00μs
CH2 ∫0.40mV

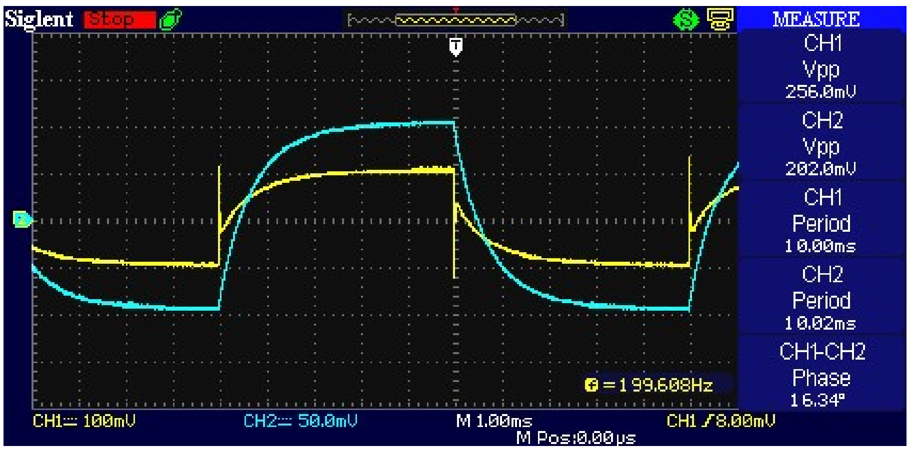

Siglent
Trig'd
MEASURE
CH1
Vpp
74.00mV
CH2
Vpp
52.00mV
CH1
Period
999.2μs
CH2
Period
1.00ms
CH1-CH2
Phase
56.20°
=1.00000KHz
CH1⎓ 50.0mV
CH2⎓ 50.0mV
M 250μs
M Pos:0.00μs
CH2 ∫2.00mV

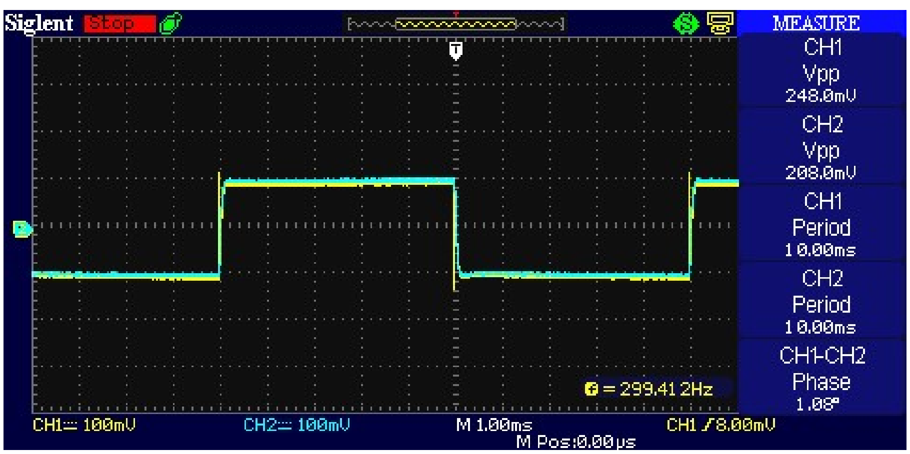

Siglent Trig'd
MEASURE
CH1
Vpp
100.0mV
CH2
Vpp
88.00mV
CH1
Period
2.00ms
CH2
Period
2.00ms
CH1-CH2
Phase
32.76°
= 500.000Hz
CH1 50.0mV
CH2 50.0mV
M 1.00ms
M Pos:0.00μs
CH2 2.00mV

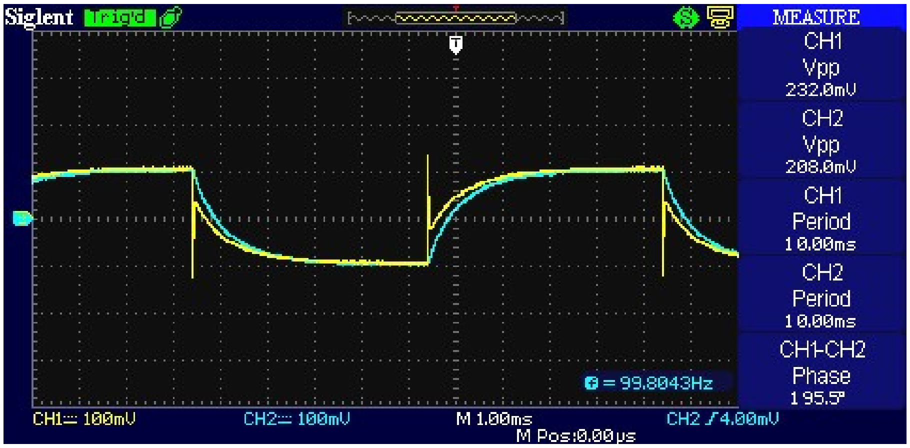

Siglent Trig'd
MEASURE
CH1
Vpp
156.0mV
CH2
Vpp
152.0mV
CH1
Period
5.00ms
CH2
Period
5.00ms
CH1-CH2
Phase
15.85°
Store Data Success!
= 199.608Hz
CH1 50.0mV
CH2 50.0mV
M 1.00ms
M Pos:0.00μs
CH2 2.00mV

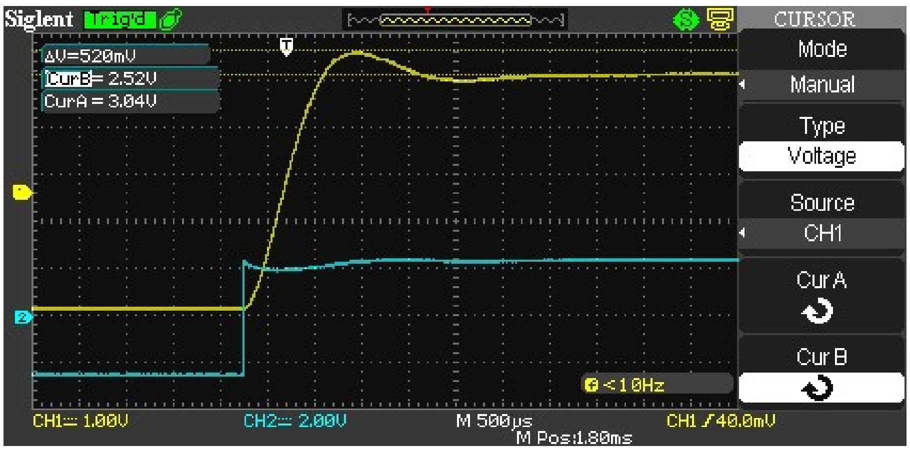

Siglent Armed
MEASURE
CH1
Vpp
264.0mV
CH2
Vpp
186.0mV
CH1
Period
10.00ms
CH2
Period
10.00ms
CH1-CH2
Phase
1.58°
= 100.782Hz
CH1 100mV
CH2 50.0mV
M 2.50ms
M Pos:0.00μs
CH2 -20.0mV

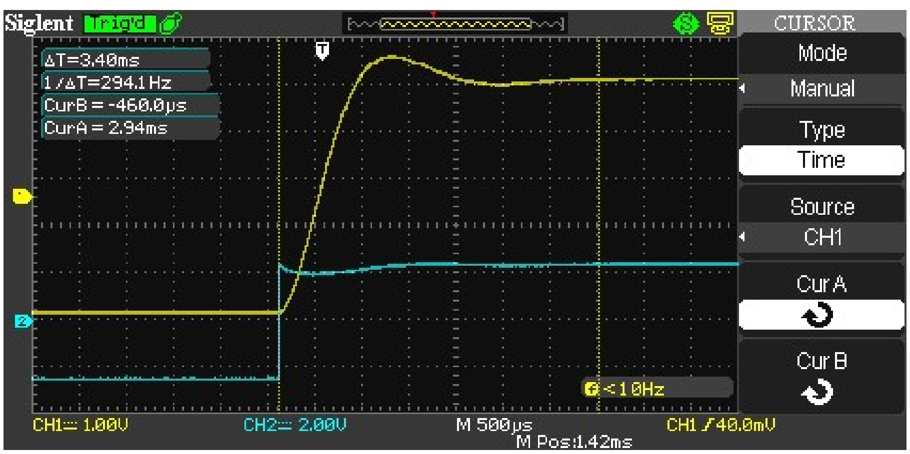

Siglent Stop
MEASURE
CH1 Vpp 264.0mV
CH2 Vpp 188.0mV
CH1 Period 10.00ms
CH2 Period 10.00ms
CH1-CH2 Phase 1.73°
= 99.8043Hz
CH1 100mV
CH2 50.0mV
M 1.00ms
M Pos:0.00μs
CH1 8.00mV

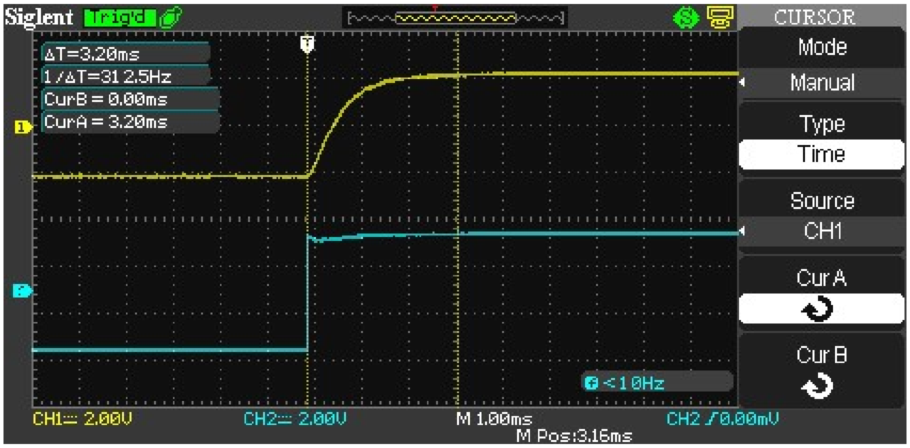

Siglent Stop
MEASURE
CH1 Vpp 256.0mV
CH2 Vpp 202.0mV
CH1 Period 10.00ms
CH2 Period 10.02ms
CH1-CH2 Phase 16.34°
= 199.608Hz
CH1 100mV
CH2 50.0mV
M 1.00ms
M Pos:0.00μs
CH1 8.00mV

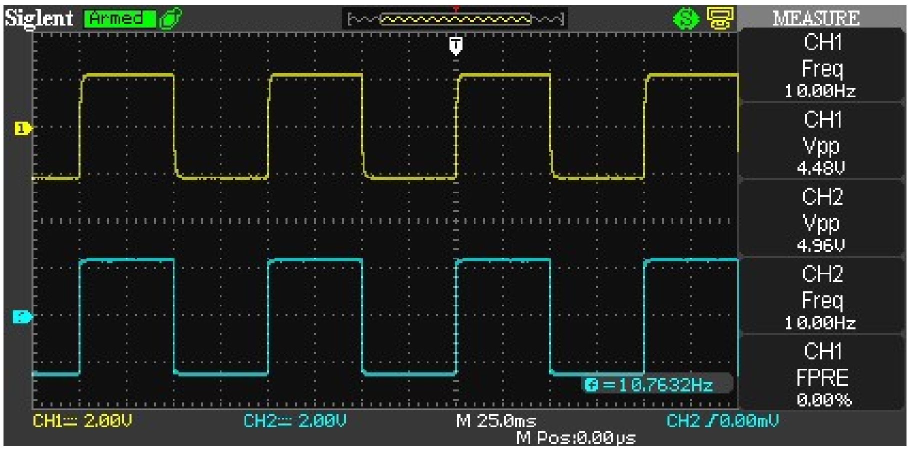

Siglent Stop
MEASURE
CH1 Vpp 248.0mV
CH2 Vpp 208.0mV
CH1 Period 10.00ms
CH2 Period 10.00ms
CH1-CH2 Phase 1.08°
= 299.412Hz
CH1 100mV
CH2 100mV
M 1.00ms
M Pos:0.00μs
CH1 8.00mV

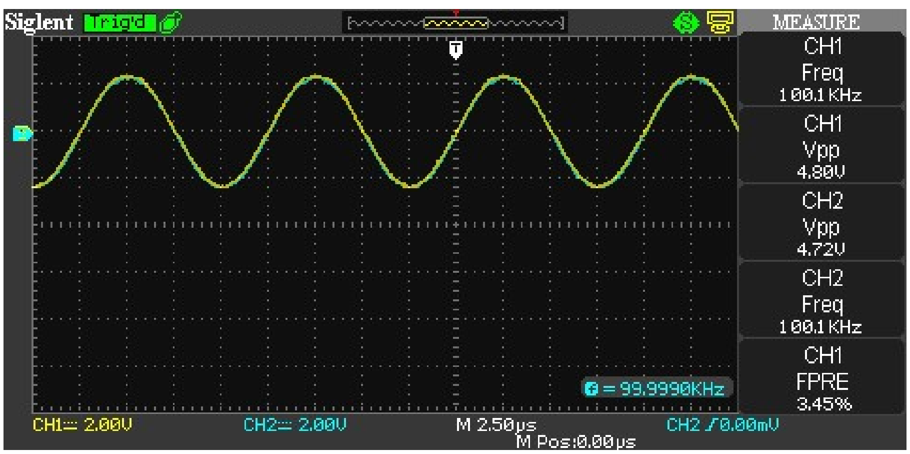

Siglent
Trig'd
MEASURE
CH1
Vpp
232.0mV
CH2
Vpp
208.0mV
CH1
Period
10.00ms
CH2
Period
10.00ms
CH1-CH2
Phase
195.5°
= 99.8043Hz
CH1 100mV
CH2 100mV
M 1.00ms
M Pos:0.00μs
CH2 4.00mV

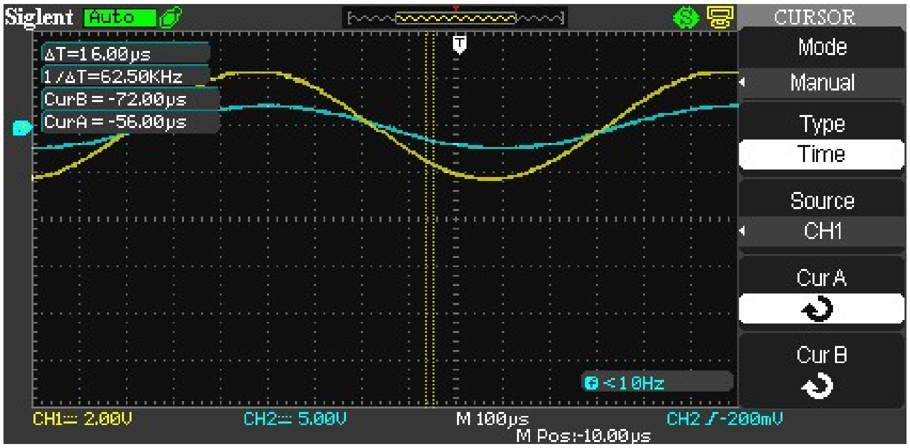

Siglent
Trig'd
CURSOR
ΔV=520mV
CurB= 2.52V
CurA = 3.04V
Mode
Manual
Type
Voltage
Source
CH1
Cur A
Cur B
< 10Hz
CH1 1.00V
CH2 2.00V
M 500μs
M Pos:1.80ms
CH1 40.0mV

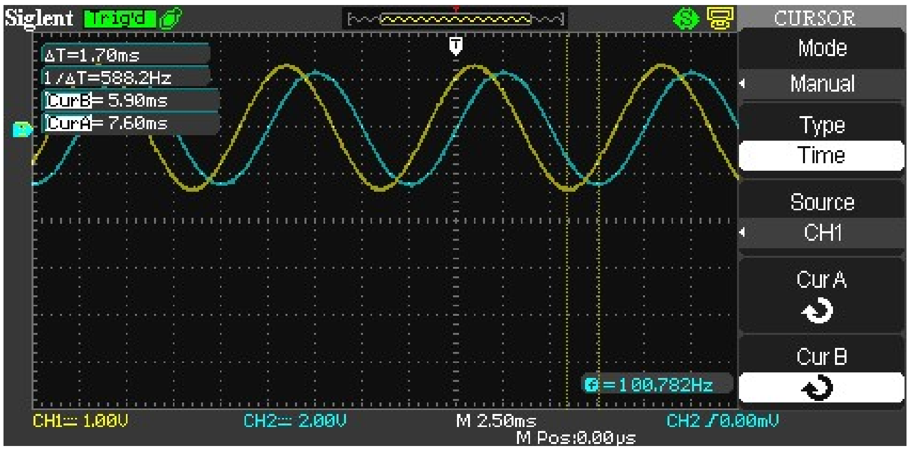

Siglent
Trig'd
CURSOR
ΔT=3.40ms
1/ΔT=294.1Hz
CurB = -460.0μs
CurA = 2.94ms
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
< 10Hz
CH1 1.00V
CH2 2.00V
M 500μs
M Pos:1.42ms
CH1 40.0mV

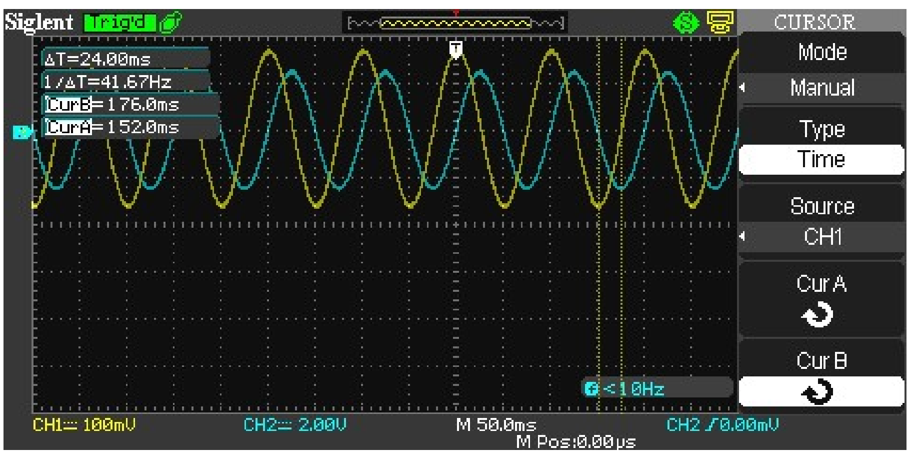

Siglent Trig'd
CURSOR
ΔT=3.20ms
1/ΔT=312.5Hz
CurB=0.00ms
CurA=3.20ms
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
<10Hz
CH1 2.00V
CH2 2.00V
M 1.00ms
M Pos:3.16ms
CH2 0.00mV

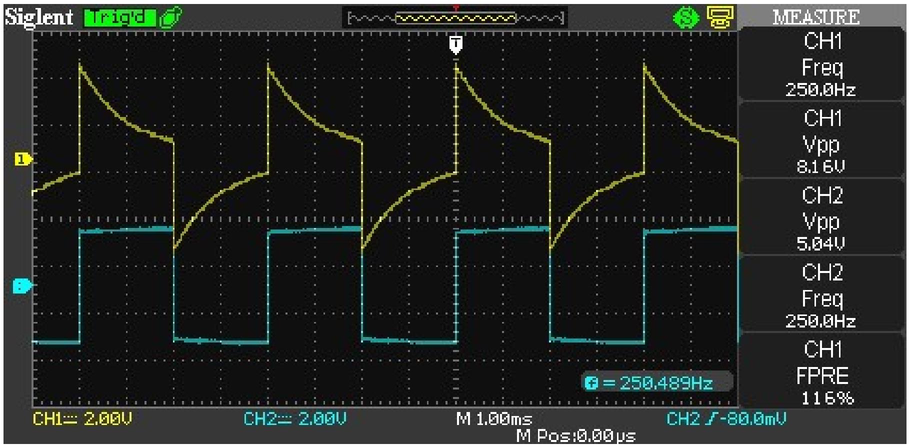

Siglent Armed
MEASURE
CH1
Freq
10.00Hz
CH1
Vpp
4.48V
CH2
Vpp
4.96V
CH2
Freq
10.00Hz
CH1
FPRE
0.00%
=10.7632Hz
CH1 2.00V
CH2 2.00V
M 25.0ms
M Pos:0.00μs
CH2 0.00mV

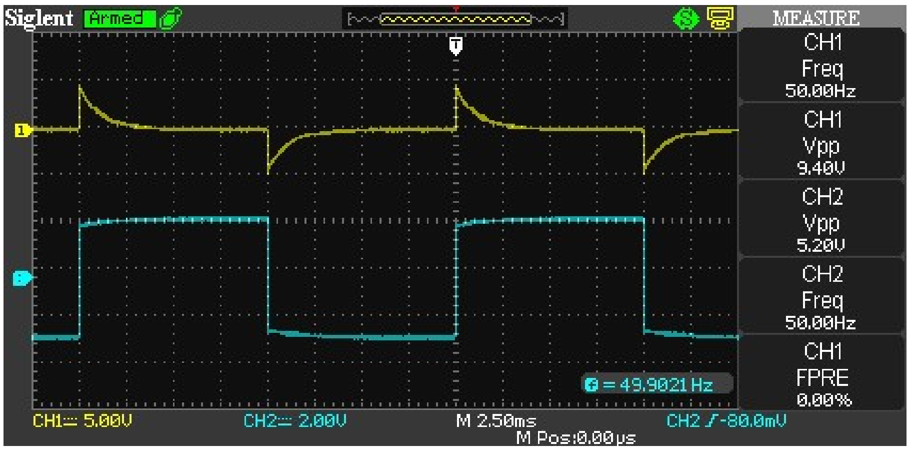

Siglent Trig'd
MEASURE
CH1
Freq
100.1KHz
CH1
Vpp
4.80V
CH2
Vpp
4.72V
CH2
Freq
100.1KHz
CH1
FPRE
3.45%
=99.9990KHz
CH1 2.00V
CH2 2.00V
M 2.50μs
M Pos:0.00μs
CH2 0.00mV

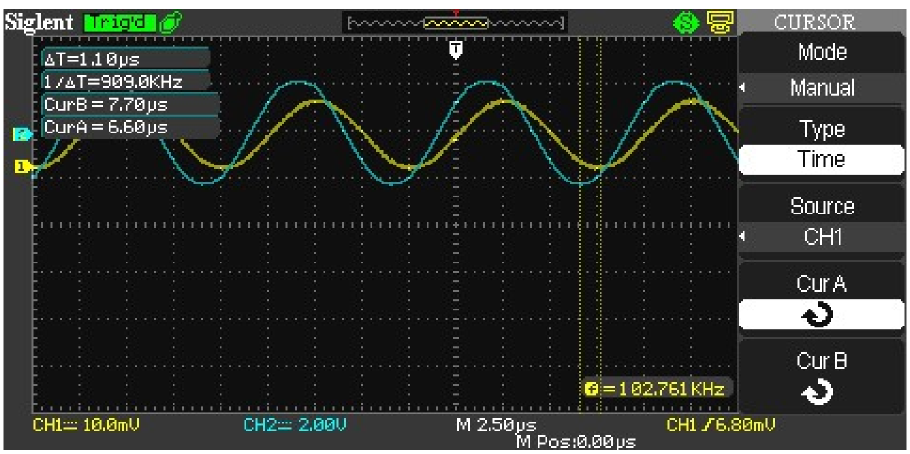

Siglent Auto
ΔT=16.00μs
1/ΔT=62.50KHz
CurB=-72.00μs
CurA=-56.00μs
<10Hz
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
CH1 2.00V
CH2 5.00V
M 100μs
M Pos:-10.00μs
CH2 -200mV

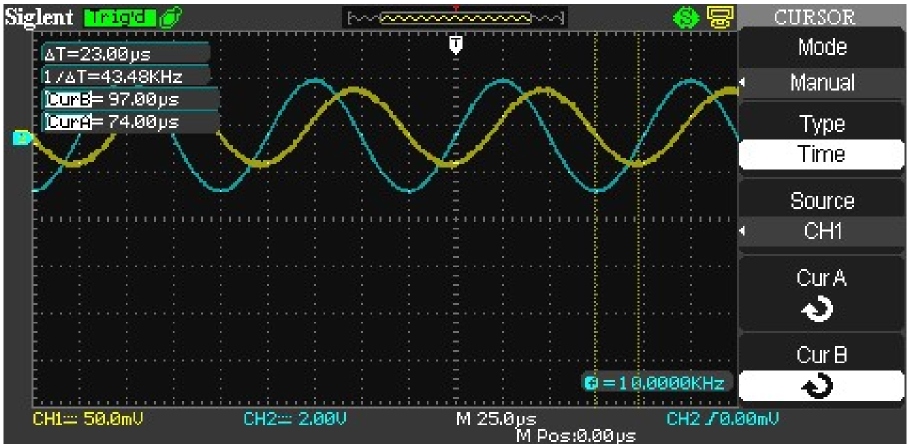

Siglent Trig'd
ΔT=1.70ms
1/ΔT=588.2Hz
CurB= 5.90ms
CurA= 7.60ms
=100.782Hz
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
CH1 1.00V
CH2 2.00V
M 2.50ms
M Pos:0.00μs
CH2 0.00mV

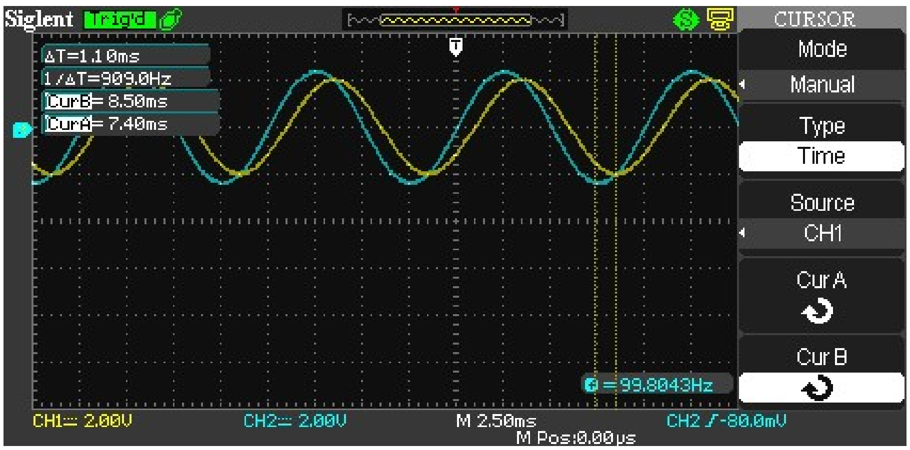

Siglent Trig'd
ΔT=24.00ms
1/ΔT=41.67Hz
CurB= 176.0ms
CurA= 152.0ms
<10Hz
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
CH1 100mV
CH2 2.00V
M 50.0ms
M Pos:0.00μs
CH2 0.00mV

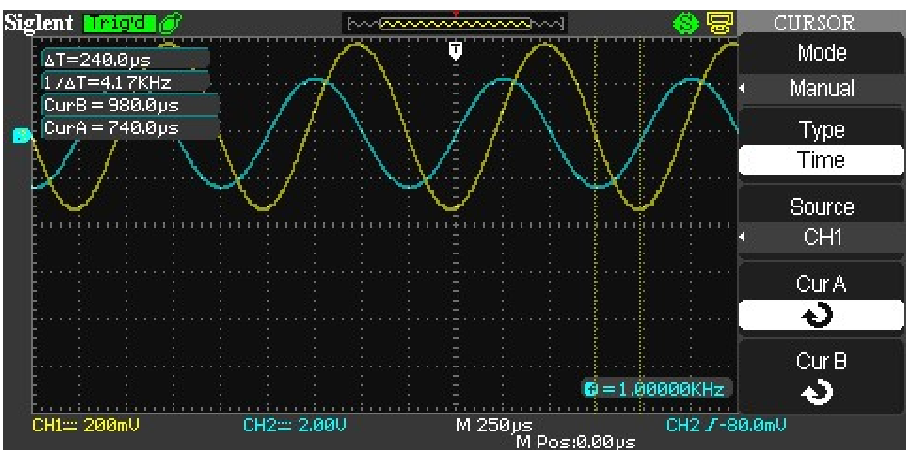

Siglent
Trig'd
MEASURE
CH1
Freq
250.0Hz
CH1
Vpp
8.16V
CH2
Vpp
5.04V
CH2
Freq
250.0Hz
CH1
FPRE
11.6%
= 250.489Hz
CH1 2.00V
CH2 2.00V
M 1.00ms
M Pos:0.00μs
CH2 -80.0mV

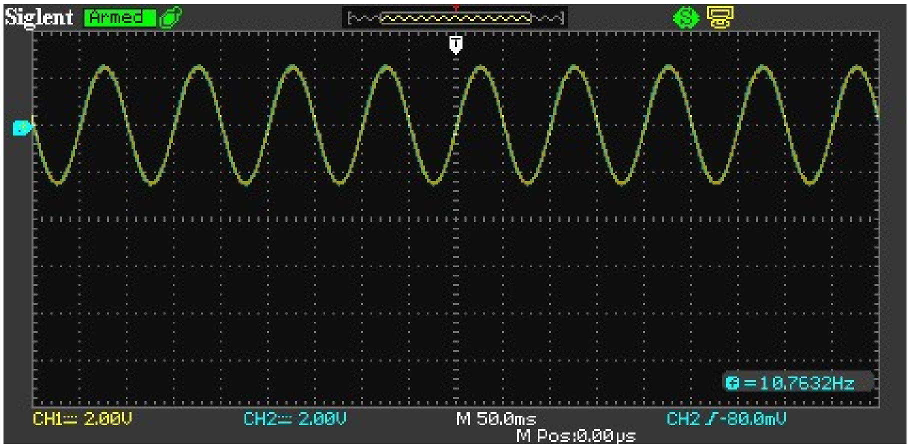

Siglent
Armed
MEASURE
CH1
Freq
50.00Hz
CH1
Vpp
9.40V
CH2
Vpp
5.20V
CH2
Freq
50.00Hz
CH1
FPRE
0.00%
= 49.9021Hz
CH1 5.00V
CH2 2.00V
M 2.50ms
M Pos:0.00μs
CH2 -80.0mV

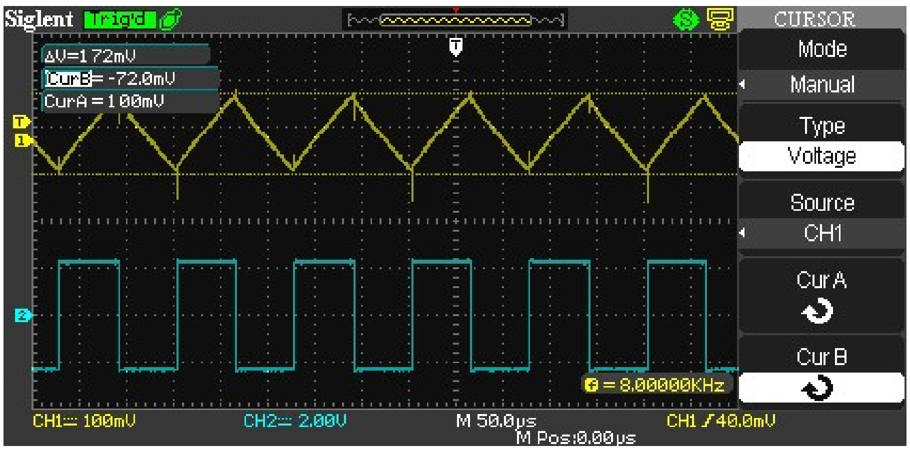

Siglent
Trig'd
CURSOR
ΔT=1.10μs
1/ΔT=909.0KHz
CurB = 7.70μs
CurA = 6.60μs
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
= 102.761KHz
CH1 10.0mV
CH2 2.00V
M 2.50μs
M Pos:0.00μs
CH1 6.80mV

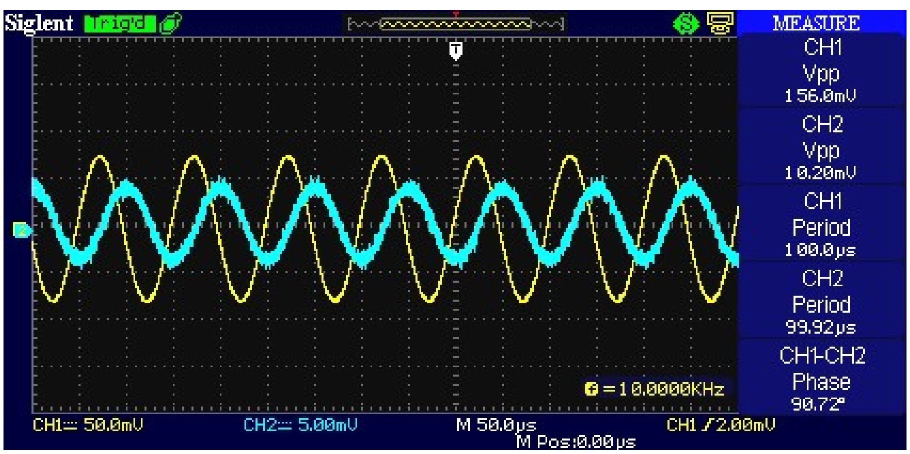

Siglent
Trig'd
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
ΔT=23.00μs
1/ΔT=43.48KHz
CurB= 97.00μs
CurA= 74.00μs
=10.0000KHz
CH1 50.0mV
CH2 2.00V
M 25.0μs
M Pos:0.00μs
CH2 0.00mV

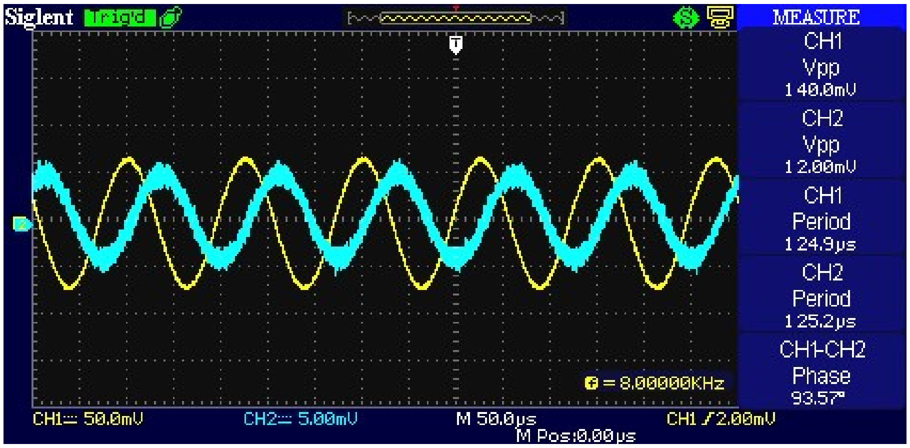

Siglent
Trig'd
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
ΔT=1.10ms
1/ΔT=909.0Hz
CurB= 8.50ms
CurA= 7.40ms
=99.8043Hz
CH1 2.00V
CH2 2.00V
M 2.50ms
M Pos:0.00μs
CH2 -80.0mV

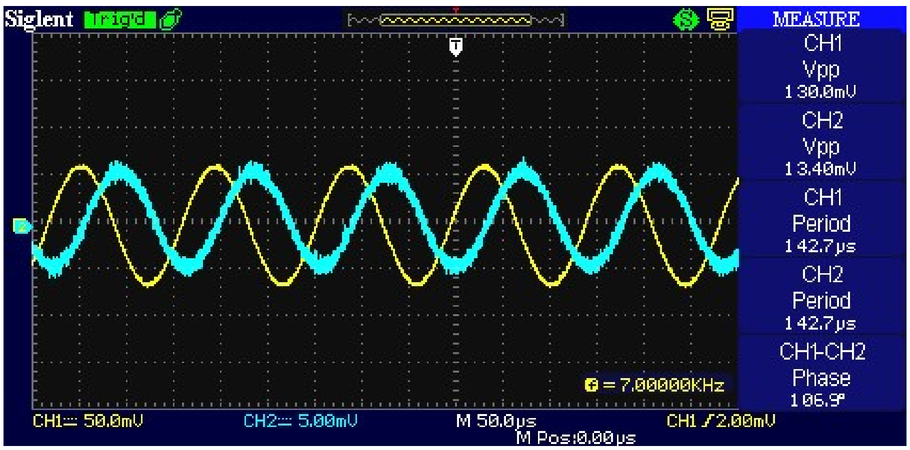

Siglent
Trig'd
CURSOR
Mode
Manual
Type
Time
Source
CH1
Cur A
Cur B
ΔT=240.0μs
1/ΔT=4.17KHz
CurB = 980.0μs
CurA = 740.0μs
=1.00000KHz
CH1 200mV
CH2 2.00V
M 250μs
M Pos:0.00μs
CH2 -80.0mV

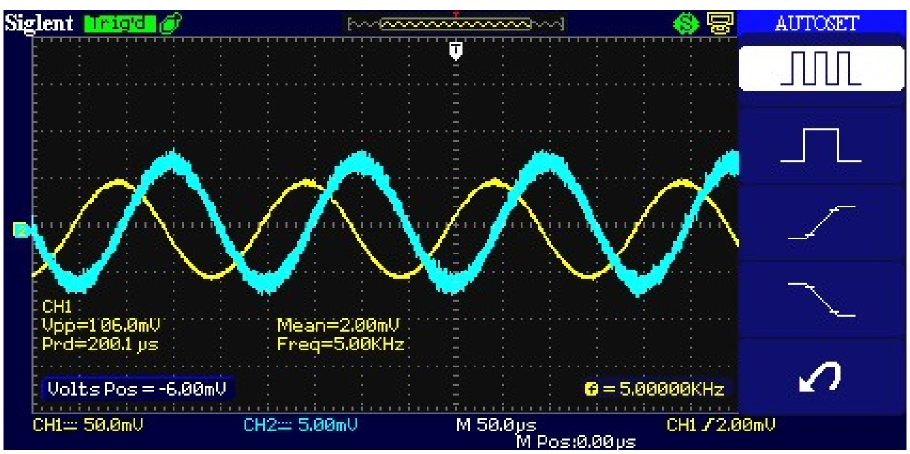

Siglent
Armed
=10.7632Hz
CH1 2.00V
CH2 2.00V
M 50.0ms
M Pos:0.00μs
CH2 -80.0mV

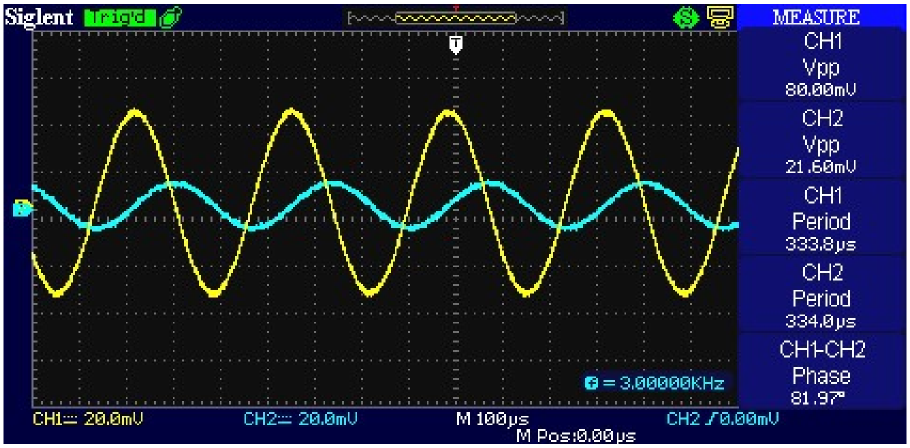

Siglent
Trig'd
ΔV=172mV
CurB= -72.0mV
CurA=100mV
CURSOR
Mode
Manual
Type
Voltage
Source
CH1
Cur A
Cur B
=8.00000KHz
CH1 100mV
CH2 2.00V
M 50.0μs
M Pos:0.00μs
CH1 40.0mV

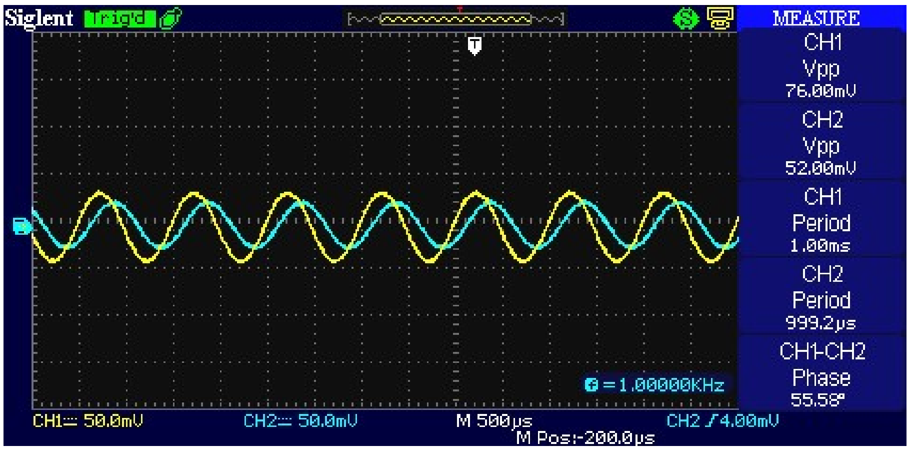

Siglent
Trig'd
MEASURE
CH1
Vpp
156.0mV
CH2
Vpp
10.20mV
CH1
Period
100.0μs
CH2
Period
99.92μs
CH1-CH2
Phase
90.72°
=10.0000KHz
CH1 50.0mV
CH2 5.00mV
M 50.0μs
M Pos:0.00μs
CH1 2.00mV

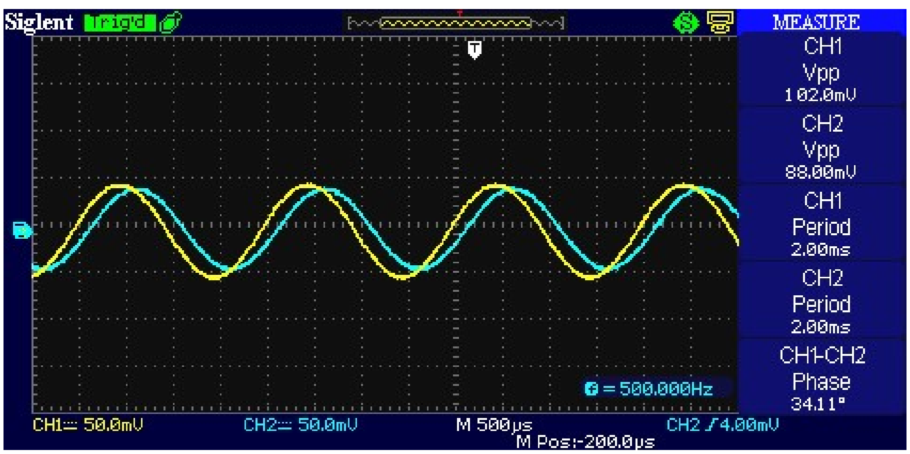

Siglent
Trig'd
MEASURE
CH1
Vpp
140.0mV
CH2
Vpp
12.00mV
CH1
Period
124.9μs
CH2
Period
125.2μs
CH1-CH2
Phase
93.57°
= 8.00000KHz
CH1 50.0mV
CH2 5.00mV
M 50.0μs
M Pos:0.00μs
CH1 2.00mV

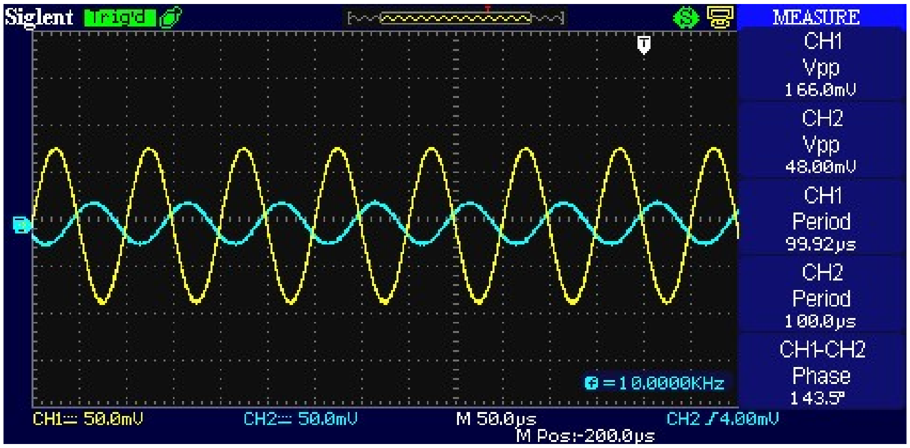

Siglent
Trig'd
MEASURE
CH1
Vpp
130.0mV
CH2
Vpp
13.40mV
CH1
Period
142.7μs
CH2
Period
142.7μs
CH1-CH2
Phase
106.9°
= 7.00000KHz
CH1 50.0mV
CH2 5.00mV
M 50.0μs
M Pos:0.00μs
CH1 2.00mV

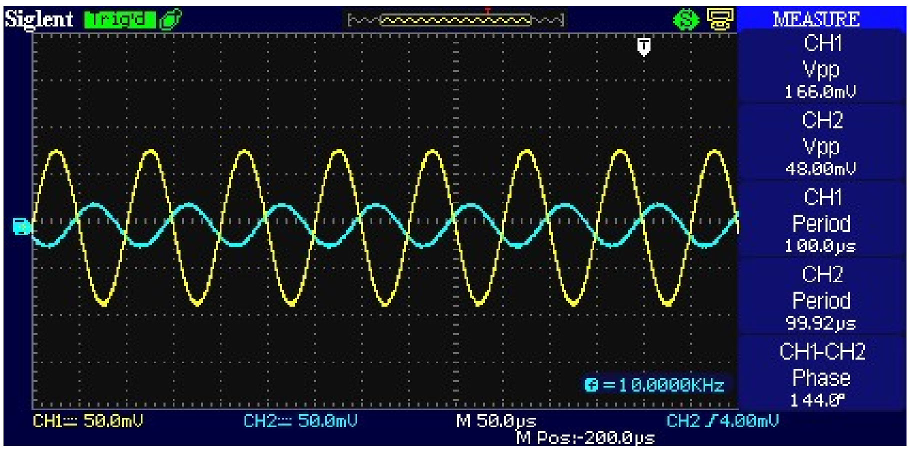

Siglent
Trig'd
AUTOSET
CH1
Vpp=106.0mV
Prd=200.1 μs
Mean=2.00mV
Freq=5.00KHz
Volts Pos = -6.00mV
= 5.00000KHz
CH1 50.0mV
CH2 5.00mV
M 50.0μs
M Pos:0.00μs
CH1 2.00mV

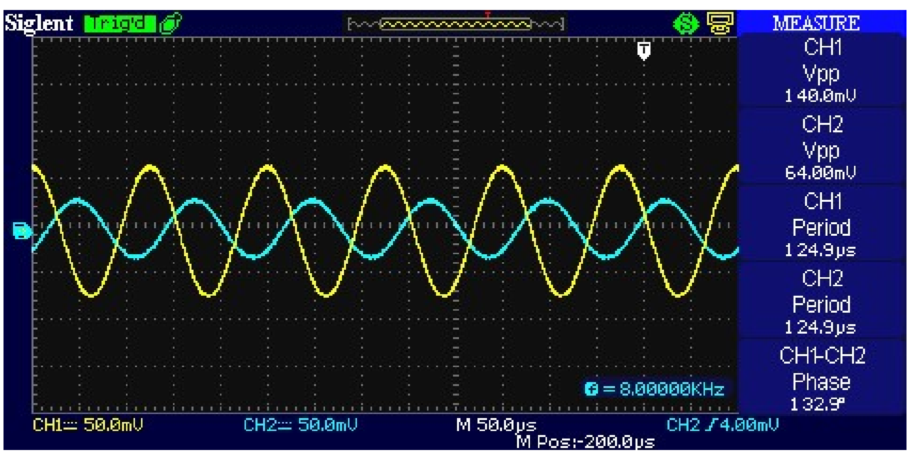

Siglent
Trig'd
MEASURE
CH1
Vpp
80.00mV
CH2
Vpp
21.60mV
CH1
Period
333.8µs
CH2
Period
334.0µs
CH1-CH2
Phase
81.97°
= 3.00000KHz
CH1 20.0mV
CH2 20.0mV
M 100µs
M Pos:0.00µs
CH2 0.00mV

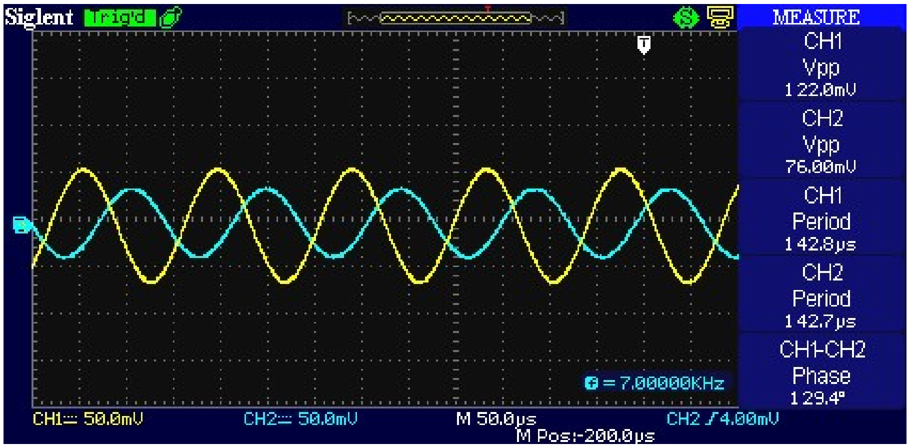

Siglent
Trig'd
MEASURE
CH1
Vpp
76.00mV
CH2
Vpp
52.00mV
CH1
Period
1.00ms
CH2
Period
999.2µs
CH1-CH2
Phase
55.58°
= 1.00000KHz
CH1 50.0mV
CH2 50.0mV
M 500µs
M Pos:-200.0µs
CH2 4.00mV

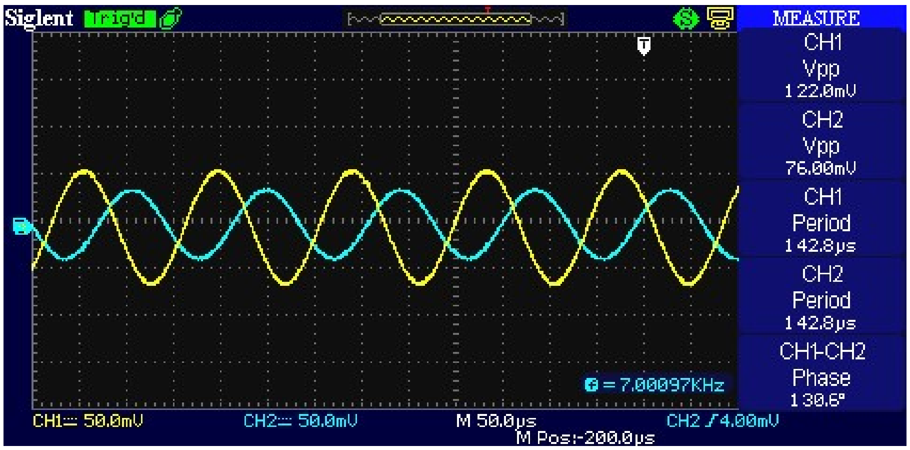

Siglent
Trig'd
MEASURE
CH1
Vpp
102.0mV
CH2
Vpp
88.00mV
CH1
Period
2.00ms
CH2
Period
2.00ms
CH1-CH2
Phase
34.11°
= 500.000Hz
CH1 50.0mV
CH2 50.0mV
M 500µs
M Pos:-200.0µs
CH2 4.00mV


Siglent
Trig'd
MEASURE
CH1
Vpp
166.0mV
CH2
Vpp
48.00mV
CH1
Period
99.92µs
CH2
Period
100.0µs
CH1-CH2
Phase
143.5°
=10.0000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0µs
M Pos:-200.0µs
CH2 4.00mV


Siglent
Trig'd
MEASURE
CH1
Vpp
166.0mV
CH2
Vpp
48.00mV
CH1
Period
100.0µs
CH2
Period
99.92µs
CH1-CH2
Phase
144.0°
=10.0000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0µs
M Pos:-200.0µs
CH2 4.00mV


Siglent
Trig'd
MEASURE
CH1
Vpp
140.0mV
CH2
Vpp
64.00mV
CH1
Period
124.9µs
CH2
Period
124.9µs
CH1-CH2
Phase
132.9°
=8.00000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0µs
M Pos:-200.0µs
CH2 4.00mV


Siglent
Trig'd
MEASURE
CH1
Vpp
122.0mV
CH2
Vpp
76.00mV
CH1
Period
142.8μs
CH2
Period
142.7μs
CH1-CH2
Phase
129.4°
= 7.00000KHz
CH1 50.0mV
CH2 50.0mV
M 50.0μs
M Pos:-200.0μs
CH2 4.00mV


Siglent
Trig'd
MEASURE
CH1
Vpp
122.0mV
CH2
Vpp
76.00mV
CH1
Period
142.8μs
CH2
Period
142.8μs
CH1-CH2
Phase
130.6°
= 7.00097KHz
CH1 50.0mV
CH2 50.0mV
M 50.0μs
M Pos:-200.0μs
CH2 4.00mV